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Structuring scientific knowledge for smart materials and 4D printing: An ontology enrichment framework using retrieval-augmented multimodal large language models

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ABSTRACT

4D printing – a rapidly evolving field that integrates additive manufacturing with stimuli-responsive materials – enables the creation of structures that change shape or properties over time. As the field advances, it poses significant challenges in synthesizing cross-disciplinary knowledge spanning manufacturing, materials science, chemistry, physics, engineering, and design. To address this, we propose a novel framework that leverages multimodal large language models and retrieval-augmented generation for ontology learning tailored to the 4D printing domain. This framework systematically extracts, maps, and interconnects insights from diverse sources, including scientific literature and experimental datasets, therefore massively enriching and populating a comprehensive multi-domain ontology. This enables the construction of interconnected knowledge graphs that capture relationships among material compositions, properties, stimuli interactions, process parameters, and design strategies. By bridging semantic gaps across involved disciplines, the framework enhances interpretability, supports informed material selection and distribution, and accelerates innovation in the development of programmable and multifunctional structures by 4D printing. Overall, this integrative approach represents a foundational step toward structuring a rapidly growing body of scientific efforts to enable advanced active materials and intelligent structures.

1. Introduction

The emergence of 4D printing – a technology combining smart materials and additive manufacturing (AM) – has opened new frontiers in fields requiring adaptive, deployable, or transformative structures (Demoly and André, 2022). This paradigm enables objects to self-transform in response to external stimuli such as heat, light, moisture, solvent, or magnetic/electric fields (Tibbits, 2013; Ge et al., 2013). The scientific landscape of 4D printing is both rapidly evolving and inherently multidisciplinary, encompassing fields such as materials science, chemistry, mechanical engineering, process engineering, and biomimicry (Demoly et al., 2021). Since its inception in 2013, the field has experienced exponential growth, with more than 3500 publications and an estimated annual growth rate of approximately 40%, according to the Web of Science database (Demoly and André, 2021b,a, 2024). Key challenges in advancing 4D printing include improving the printability of smart materials, enhancing their mechanical and actuation performance, promoting safe and sustainable deployment, and ensuring reliability under cyclic stimuli and real-world conditions (Demoly et al., 2021). These challenges can be considered as interdependent, especially when designing and developing practical 4D-printed systems, where trade-offs between material properties, process parameters, and functional requirements must be carefully balanced (Demoly et al., 2021). While this diversity has accelerated innovation, it has also led to fragmented and rapidly evolving bodies of knowledge, therefore making it increasingly difficult to organize, reuse, and reason over existing research in a coherent manner.

In such a context, ontologies provide a formalized approach to capturing domain-specific knowledge. Several efforts have attempted to address knowledge organization in materials science through ontologies, including European Materials Modeling Ontology (EMMO) (Ghedini et al., 2017), Materials Design Ontology (MDO) (Li et al., 2020), Materials Ontology (MatOnto) (Cheung et al., 2008), Materials

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Web Ontology Language ontology (MatOWL) (Zhang et al., 2009), NanoParticle Ontology (Thomas et al., 2011), eNanoMapper ontology (Hastings et al., 2015), Metallic Materials Ontology (MMOY), derived from Yet Another Great Ontology (YAGO) (Zhang et al., 2016), Materials and Molecules Basic Ontology (MAMBO) (Piane et al., 2021), Dislocation Ontology (Ihsan et al., 2021), and Platform Material Digital ontology (PMD) (Alam et al., 2021). In particular, MDO enables semantic querying across diverse materials databases like the Materials Project (Jain et al., 2013), the Open Quantum Materials Database (OQMD) (Saal et al., 2013), the Novel Materials Discovery (NOMAD) (Draxl and Scheffler, 2018), and the Automatic Flow Library for Materials Discovery (AFLOWLIB) (Curtarolo et al., 2012), thus offering a unified framework for navigating the vast landscape of materials data. These ontologies provide structured, interpretable representations and support semantic querying across heterogeneous materials database. However, they remain largely static, require manual knowledge engineering, and struggle to adapt to fast-growing, interdisciplinary domains such as 4D printing. Among ongoing initiatives in material formalization (Lambrix et al., 2024) and AM knowledge capture (Jee and Witherell, 2017; Dinar and Rosen, 2017), taxonomic classifications (Solórzano-Requejo et al., 2025) and design-oriented ontologies such as HERMES – an ontology integrates 3- and 4-dimensionalism theories to ensure semantic coherence across the involved disciplines and abstraction levels – (Dimassi et al., 2021b, 2023), offer valuable conceptual foundations but remain limited in scalability, automation, and continuous knowledge update. Consequently, no existing ontology adequately supports the dynamic and cross-disciplinary knowledge evolution required in 4D printing. This limitation is not only related to the conceptual coverage of existing ontologies, but also to the difficulty of continuously updating them from rapidly expanding scientific literature, heterogeneous datasets, and multimodal technical evidence.

To complement these structured but mostly static knowledge representations, artificial intelligence (AI) methods offer new opportunities for extracting, aligning, and updating domain knowledge from unstructured and multimodal sources. In particular, natural language processing (NLP) and machine learning (ML) techniques can support the automatic identification of concepts, relations, and evidence from scientific texts, figures, and datasets (Li, 2018). In this context, large language models (LLMs), such as GPT (Radford et al., 2019) and BERT (Devlin et al., 2019), have transformed NLP by enabling advanced capabilities in translation, question answering, and text generation (Zhao et al., 2023; Xu and Poo, 2023). Built on the Transformer architecture (Vaswani, 2017), these models leverage vast datasets to achieve high levels of contextual understanding. Scaling LLMs has led to emergent reasoning capabilities, including in-context learning (ICL) (Peng et al., 2023), chain-of-thought (CoT) (Wei et al., 2022), and retrieval-augmented generation (RAG) (Gao et al., 2023). These advances mitigate some limitations of conventional AI models by enabling real-time knowledge retrieval and contextual inference. Additionally, the recent development of multimodal LLMs (MLLMs), such as GPT-4V (Achiam et al., 2023) and Gemini (Team et al., 2023), has further expanded AI's ability to integrate textual, visual, and symbolic information (Yin et al., 2024). These models are particularly relevant for domains like 4D printing, where information must be captured across modalities to support design synthesis.

Through these advancements, LLMs continue to face fundamental limitations in terms of interpretability and domain specificity. Their probabilistic generation process can result in hallucinations and unreliable outputs, which is particularly problematic in specialized and safety-critical domains such as 4D printing (Ji et al., 2023). This contrast highlights a key research gap: ontologies offer rigor, interpretability, and logical consistency, while LLMs provide scalability and adaptability, but existing approaches rarely integrate both in a principled and domain-specific manner. Current ontology learning and enrichment methods either rely heavily on manual intervention or adopt fully automated pipelines that risk semantic drift, bias propagation, and error accumulation (Hogan et al., 2022; Maedche and Staab, 2001; Cimiano and Völker, 2005). Traditional manual annotation remains the most reliable option, yet it is labor-intensive, subjective, and unscalable when dealing with large-scale or rapidly evolving datasets. To address this, semi-automatic approaches such as Phrase2Onto (Pour et al., 2023) have been proposed. This method applies phrase-level topic modeling to scientific corpora in order to identify recurring linguistic patterns that may correspond to new ontology concepts. Candidate terms are then suggested to experts for integration into the ontology. While effective in surfacing domain-specific terminology, Phrase2Onto still requires significant human input for validation and disambiguation, which can introduce subjectivity and limit scalability. In parallel, fully automated methods using NLP and ML (e.g., Text2Onto) expedite ontology population by extracting entities and relations directly from text (Cimiano and Völker, 2005). Yet, these approaches remain highly dependent on the quality of training data, raising risks of bias propagation, semantic drift, or shallow representations when applied to domain-specific corpora (Gallegos and coauthors, 2024). More recent text-to-knowledge-graph (text2KG) approaches attempt to construct knowledge graphs (KGs) directly from unstructured text, integrating steps such as entity recognition, relation extraction, canonicalization, and ontology alignment (Hogan et al., 2022; Jing et al., 2025). While these pipelines improve scalability, they often suffer from error accumulation across stages and lack mechanisms for continuous validation. Emerging solutions seek to reduce these limitations. For example, online clustering with LLM agents (Wu et al., 2024) allows for the incremental grouping of newly extracted knowledge units (entities, relations, or triplets) into coherent clusters without requiring extensive annotated datasets. These systems show potential for handling large-scale, continuously growing corpora, but they face challenges in maintaining logical consistency across clusters and preventing redundancy. Similarly, graph-enhanced LLM frameworks (Pan et al., 2023; Oyekan et al., 2025) attempt to embed structural constraints from knowledge graphs into the generative process, thereby improving alignment between generated triplets and ontology schemas. Finally, RAG has emerged as a promising solution (Lewis et al., 2020; Gao et al., 2023; Ludwig et al., 2025). By retrieving external domain corpora during the generation process, RAG constrains LLM outputs to be grounded in verified evidence, thereby reducing hallucinations and improving semantic relevance, as also demonstrated in recent ontology-based industrial retrieval-augmented generation systems (Ludwig et al., 2025). Combined with post-generation filtering strategies (e.g., semantic similarity checks, predicate template matching, redundancy elimination), RAG represents a scalable pathway to bridge unstructured text and structured, domain-aligned ontologies in dynamic scientific fields.

In this paper, we propose a retrieval-augmented MLLMs framework for active ontology enrichment in 4D printing. By integrating ontological structures with the generative and retrieval capabilities of MLLMs, the proposed framework enables scalable, evidence-grounded extraction of domain knowledge from both structured and unstructured sources. This hybrid approach preserves semantic rigor while allowing continuous adaptation to emerging research, thereby supporting knowledge discovery, reasoning, and reuse in the rapidly evolving field of 4D printing.

2. Methods

2.1. Description of backbone ontology for 4D printing

The rapid advancements in 4D printing have introduced a need for a structured framework to manage and formalize the diverse knowledge involved in designing transformable systems. The HERMES ontology addresses this need by providing a semantic and logical foundation for representing the dynamic behavior of 4D-printed objects (Dimassi et al., 2021b). Built upon the Basic Formal Ontology (Arp et al., 2015) and mereotopology theory (Smith, 1996), this ontology is centered on

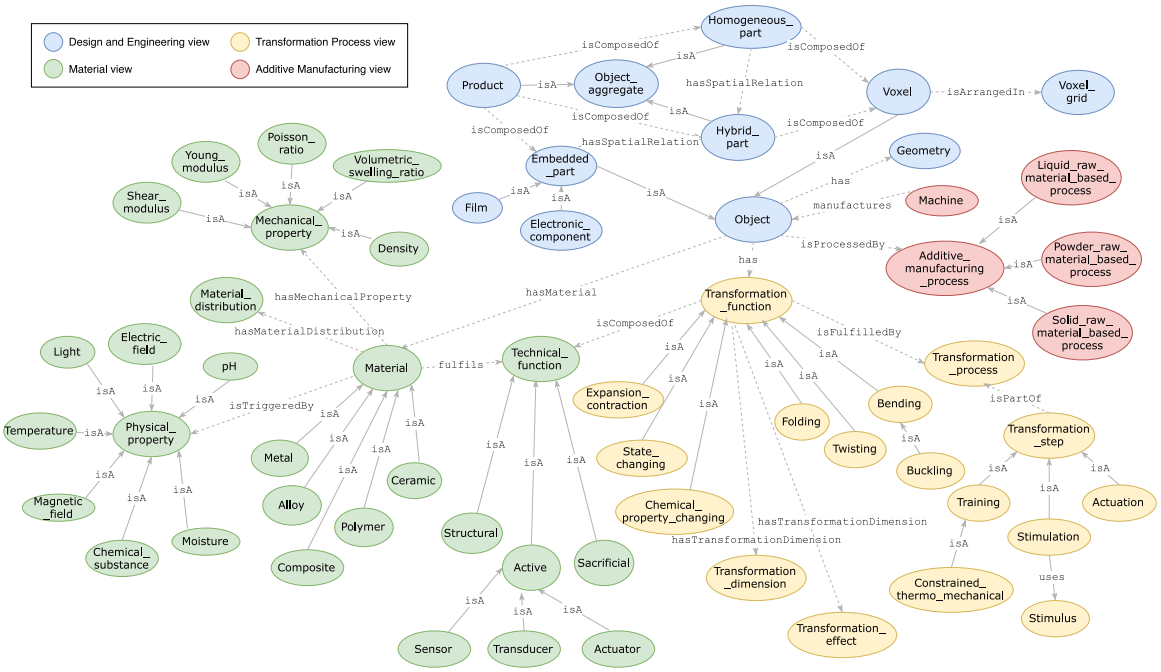


Fig. 1. Extraction of the main classes of the backbone ontology for 4D printing, structured around four main views: design and engineering, material, AM, and transformation process (Dimassi et al., 2021b) (Adapted with permission from Elsevier).

key 4D printing views, namely AM, material, transformation process, and design and engineering. These views are formalized using interconnected classes, attributes, and relationships. Fig. 1 illustrates a part of the 4D printing ontology where views are highlighted.

In this paper, the following naming conventions are used as follows: the names of classes are written in capitalized/lower case Arial, the names of attributes and relationships are capitalized/lower case Courier New, while names of instances are in italics Arial. In the design and engineering view, key classes include Object, Homogeneous_Part, which captures single-material parts, and Hybrid_Part, which integrates different materials. These elements are linked by spatiotemporal relationships to enable comprehensive modeling of complex assemblies. The material view categorizes materials based on their Technical_Function – Active, Structural, or Sacrificial – and captures their mechanical and physical properties, as well as their spatial distribution. This view is critical for selecting and designing material combinations that respond predictably to stimuli. In addition, the AM view integrates manufacturing techniques, machines, and process parameters, linking these elements to specific material behaviors. This view provides a framework for planning fabrication steps that align with the requirements of 4D printing. Finally, the transformation process view formalizes the dynamic changes that define 4D printing. It includes Transformation_function such as Bending, Twisting, and Folding, along with property changes like color or stiffness. Transformation sequences are modeled as temporal entities, enabling the logical organization of Training, Stimulation, and Actuation steps required for shape and function changes in the structures. Beyond this broad set of concepts, the ontology also incorporates individuals which are tangible entities for representing classes. For instance, the Transformation_Effect class has the following individuals, namely *One_way*, *Reversible*, and *Repeatable*. Object properties are then utilized to outline the relationships between individuals and classes, thus creating meaningful links within the ontology. Examples of these links are highlighted in dot-patterned arrows in Fig. 1. The last component of the ontology concerns data properties, which specify attributes or characteristics of the classes, such as "Folding_angle", relevant to transformation function like Folding class.

The HERMES ontology extensively utilizes description logic (DL) rules to enhance its structural and functional capabilities. These DL rules enable precise definition and restriction of classes to support sophisticated inference mechanisms. By implementing these rules, HERMES facilitates key operations such as consistency checking, which verifies the logical coherence of the ontology, as well as subsumption to determine class hierarchies, and realization to identify class memberships. The application of DL rules not only strengthens the ontology's ability to manage and classify complex knowledge effectively but also ensures that it can be interacted with and interpreted by machines. Although structured around philosophical foundations and DL rules to ensure expressivity and reasoning across abstraction levels, this ontology – like most existing material ontologies – suffers from limited capabilities for automated and large-scale learning through enrichment and population. This limitation is particularly critical in emerging and rapidly evolving research domains like 4D printing, where knowledge consolidation is essential to enhance technological readiness levels and reach practical applications.

2.2. Ontology enrichment framework

Ontology enrichment enables the enhancement of an existing preliminary ontology by automatically adding new concepts (also considered as knowledge), relationships, and individuals (meaning information or data) to make it more comprehensive and practical for a specific domain or task. To ensure both the enrichment and population of the HERMES ontology, we employ an integrated framework combining information retrieval with advanced text generation capabilities (as illustrated in Fig. 2).

Initially, we collected a curated corpus of published articles and domain-specific material datasets using targeted keywords. Each published article is split into discrete text segments and extracted figures, while each dataset table is parsed into individual records. All text and image snippets are then encoded using a fine-tuned MLLM into dense vectors and stored in a high-throughput vector index. At inference time, the LLM issues similarity queries against this index to retrieve the top-k relevant passages or images, which it incorporates as "context windows" into its prompts. From the generated and context-aware

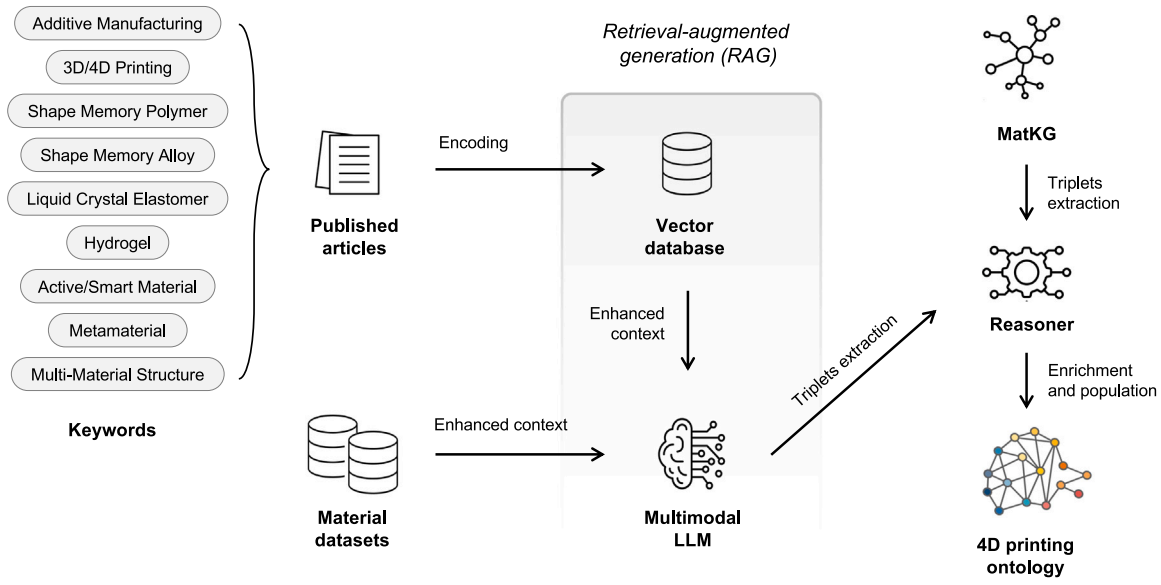


Fig. 2. Retrieval-augmented ontology construction pipeline. Keywords related to 4D printing pillars are fed into ResearchRabbit to retrieve hundreds of published articles, which are then stored in a vector database. A RAG (retrieval-augmented generation) framework combines this vector database with a MLLM that is enriched by additional datasets to extract relational triplets. The resulting triplets are then combined with triplets from a large knowledge graph database on materials science (MatKG) to populate an existing 4D printing ontology.

outputs, a downstream triplet-extraction module identifies candidate subject–predicate–object facts. These facts are merged with existing knowledge from MatKG (Venugopal and Olivetti, 2024) – one of the largest knowledge graph in materials science – and passed to a symbolic reasoner, which enforces ontology schema constraints, checks for logical consistency, and removes duplicates. Generated triplets are then translated into classes, properties, or instances, thereby populating and enriching the 4D printing ontology in a continuous loop that keeps our knowledge base both up to date and semantically rigorous.

2.3. Ontology enrichment from scientific literature

To enrich the ontology, the process starts with the selection of key terms, ie., “Additive Manufacturing”, “3D/4D Printing”, “Shape Memory Polymer”, “Shape Memory Alloy”, “Liquid Crystal Elastomer”, “Hydrogel”, “Active/Smart Material”, “Metamaterial”, and “Multi-Material Structure”. By utilizing ResearchRabbit application tool (ResearchRabbit, 2025), a scholarly publication discovery tool supported by AI, intersections among these keywords were explored to gather over 1810 published articles. The key correlations extracted are illustrated through a Venn diagram in Fig. 3.

By coupling the keywords “Additive Manufacturing” with both “3D/4D Printing” as the sole procedural node, the ontology captured the full spectrum of programmable, layer-by-layer processes, including studies involving smart materials that are not explicitly tagged “4D Printing”. By linking each major material class (“Shape Memory Polymer”, “Shape Memory Alloy”, “Liquid Crystal Elastomer”, and “Hydrogel”) to the “Active/Smart Material” category, we broadened the scope to encompass the majority of experimental work that emphasizes stimuli-responsive functionality over processing terminology. Finally, by pairing “Metamaterial” and “Multi-Material Structure” with “4D Printing”, we specifically targeted research on architectured and multi-material constructs whose geometries and properties evolve over time. The corpus of 1810 published articles spans the period 2009–2024 and covers several peer-reviewed journals in the 4D printing domain. The full list of peer-reviewed journals is provided in the Supplementary Materials.

Then, we split these published articles using tools like LLM Sherpa (Nlm-atics, 2024) for robust text extraction and semantic chunking, which divided each paper into coherent chunks based on structural elements. This approach was designed to optimize both semantic completeness and computational efficiency, ensuring that each segment retained meaningful contextual information. Chunk boundaries followed the natural discourse flow (e.g., paragraphs or logical sections) rather than fixed lengths, thereby preserving local coherence throughout the segmentation process. The Aspose tool (Aspose, 2024) was used for image extraction in order to isolate each figure into standalone image files. Each token was then embedded using BERT model and CLIP for images. This process converts the content into vector representations, allowing us to store them in a vector space, as shown in Fig. 4.

Subsequently, detailed information concerning structures, material preparation processes, and methodologies was systematically extracted from articles. Leveraging BERT embeddings, queries were encoded within a text vector space to identify the 25 nearest vectors. As illustrated in Fig. 5, selecting one text vector reveals the section highlighted in green as the most relevant to the query. This section is then utilized to identify and retrieve its corresponding relevant image through CLIP embedding model and image vector space.

By pairing each textual section with its corresponding image, we enriched the LLaVA (Large Language-and-Vision Assistant) MLLM’s input context (Parthasarathy et al., 2024; Kim et al., 2023; Jeong, 2024). The rationale for this model choice is provided in the **Supplementary Material**. This RAG process combines information retrieval with text generation, so that it helps to address challenges such as hallucination, outdated knowledge, and opaque reasoning in language models. By incorporating data from external databases, RAG ensures more accurate and credible output, particularly in knowledge-intensive tasks. This integration facilitates ongoing updates and the inclusion of specialized information, therefore making RAG a dynamic solution that combines intrinsic model knowledge with extensive external data.

Initially, we employed the few-shot learning approach (Brown et al., 2020; Hoffmann et al., 2022; Yang et al., 2022); however, this method was inefficient due to its time and memory demands related to the excessive length of contexts and often yielding imprecise results. To enhance efficiency, we fine-tuned the LLaVA model specifically for triplet extraction (Ghanem and Cruz, 2024; Liu et al., 2022; Zhang

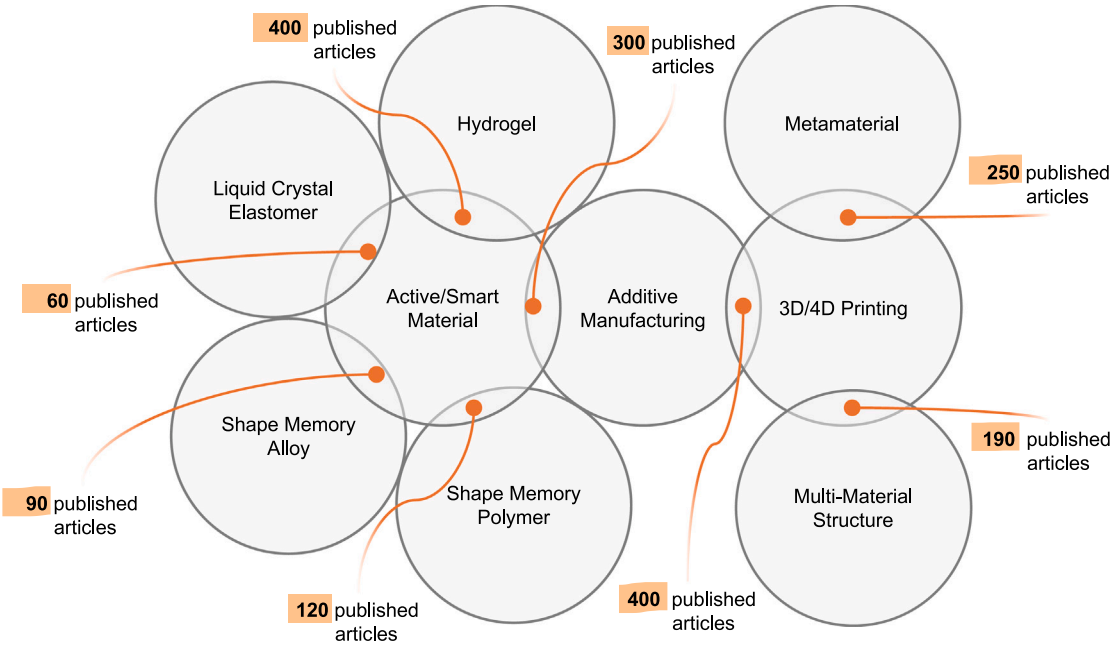


Fig. 3. Venn diagram illustrating the selected cross-domain keywords used to retrieve relevant articles published in peer-reviewed journals.

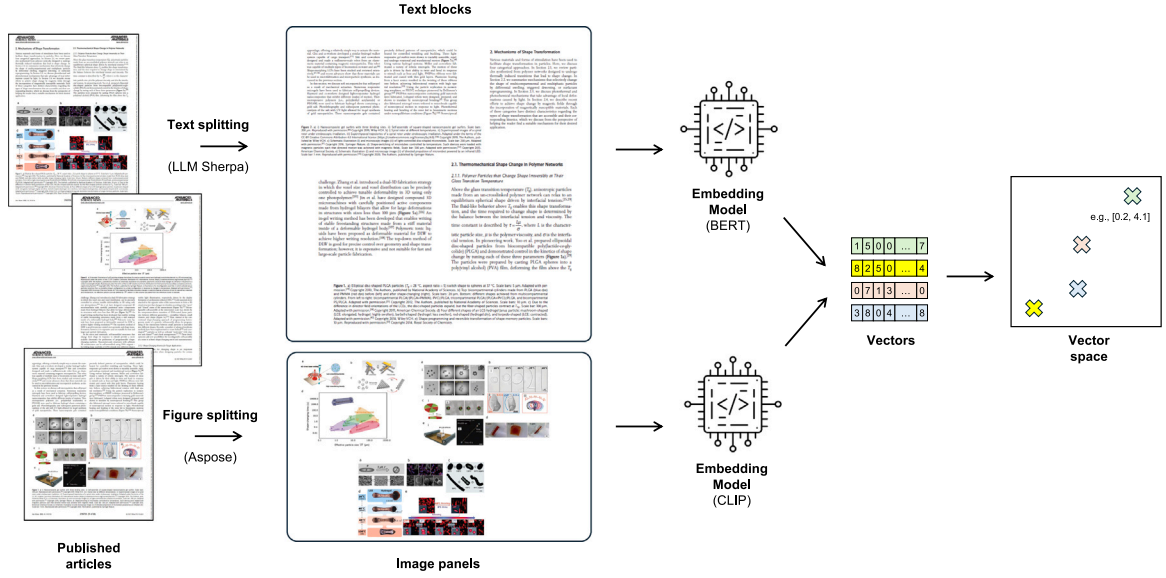


Fig. 4. Document preprocessing and multimodal embedding pipeline. Multi-page articles were first divided into text blocks (using LLM Sherpa) and figure panels (using Aspose tool). Text segments were encoded with a BERT embedding model, while image panels were encoded with CLIP. The resulting text and image embeddings were then projected into a shared vector space for downstream retrieval and similarity tasks.

et al., 2024). This process involved embedding domain-specific knowledge within the LLaVA model, thus configuring the output format appropriately and ensuring consistent performance without the need for additional tokens. The fine-tuning dataset comprised prompts, relevant images, and targeted responses, enabling the model to align more closely with our extraction logic and generate outputs in the specified format. Moreover, during fine-tuning, we trained the LLaVA model to distinguish ontology classes, object properties, data properties, and instances, while preserving the hierarchical relations among classes in accordance with ontology web language (OWL) formalism (Val-Calvo et al., 2025; Doumanas et al., 2025). This approach aims to refine the LLaVA model into a tool adept at identifying triplets specifically related to 4D printing. To construct the fine-tuned dataset, we opted for an approach to build a synthetic dataset by employing ChatGPT-4 (Achiam et al., 2023) (see Fig. 6). Relevant textual sections were extracted from the scientific corpus and paired with the most semantically aligned figure via CLIP-based cross-modal retrieval. Each image-text pair was then converted into a structured prompt comprising the raw section text, the retrieved figure, and explicit extraction constraints. In particular, ChatGPT-4 was instructed to rely exclusively on the provided sections and figures, to avoid introducing any information not supported by these inputs, and to output candidate triples strictly in a normalized subject–predicate–object format aligned with our ontology schema through controlled predicate templates and typing expectations. A one-shot demonstration example was included to stabilize the output structure and reduce format variability. Importantly, ChatGPT-4 outputs were treated as candidate annotations rather than authoritative truth: all 230 generated examples were subsequently reviewed by a

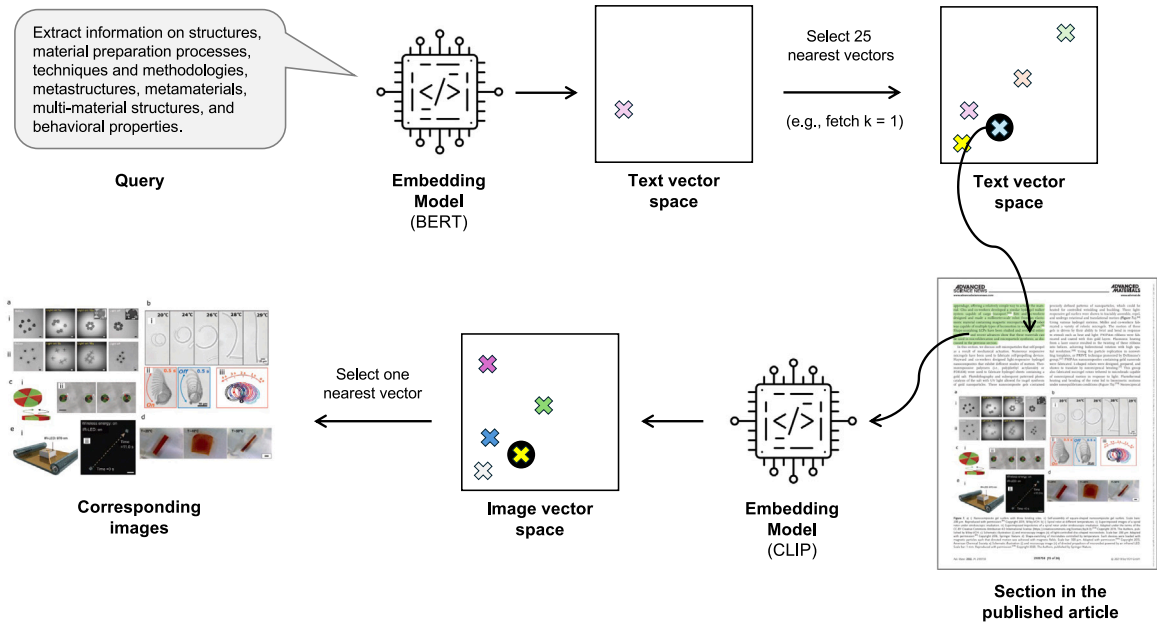


Fig. 5. Cross-modal retrieval pipeline. A user's textual query was first encoded with BERT embedding model and used to retrieve the top-25 nearest text embeddings from a document corpus. The corresponding articles were then fetched, and all embedded figures within each article were encoded using CLIP embedding model. Finally, the CLIP embeddings are compared to the original query embedding to select the single most relevant image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

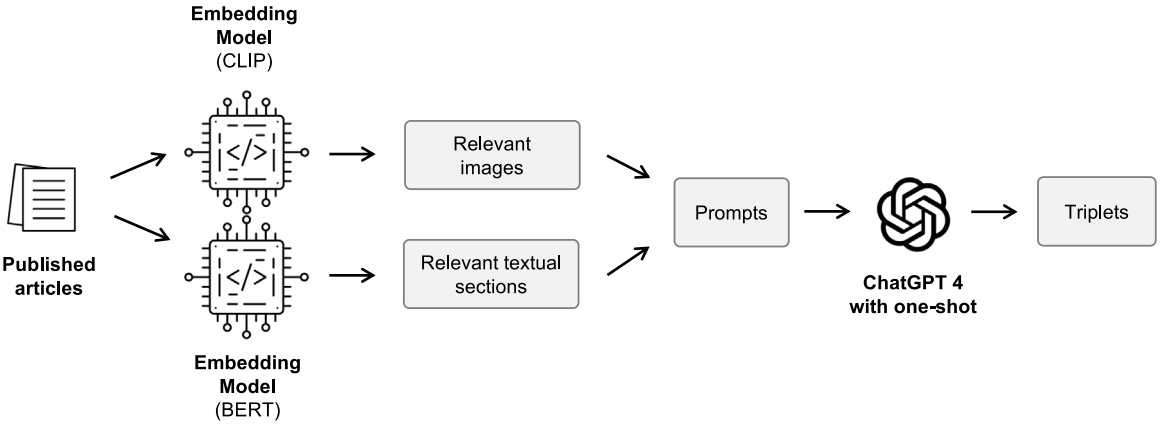


Fig. 6. Pipeline for generating the fine-tuning dataset. (1) Section selection. Relevant textual sections are automatically extracted from a corpus of scientific articles using a BERT-based semantic similarity model. (2) Image association. for each section, the most semantically aligned image is selected using CLIP-based similarity matching. (3) Generation of textual prompts. For each selected image and its corresponding section, we insert them into the prompt template. (4) Generation of triplets. Using a one-shot prompt to ChatGPT-4, convert each prompt into triplets.

4D-printing domain expert with ontology-engineering expertise, who verified semantic correctness, removed unsupported claims, corrected entity typing and relation directionality, and resolved ambiguous cases by rewriting or discarding the corresponding triples. The resulting dataset therefore constitutes an expert-curated supervision set used to fine-tune the extractor toward ontology-aligned expression and typing, rather than learning to assert facts from the model itself. The resulting dataset follows a standardized format: [prompt (containing both the section and the image), triplets], and consisted of 230 ground truth examples.

While specialized models may not always perform optimally on specific tasks, they require fine-tuning to enhance their effectiveness. Parameter efficient fine-tuning (PEFT) (Lialin et al., 2023; Ding et al., 2023; Fu et al., 2023; Xu et al., 2023; Han et al., 2024) is a technique for fine-tuning an LLM that requires fewer resources to train, making it a cost-effective option. It offers faster training times compared to traditional fine-tuning methods. By regularizing the training process, it helps maintain the model's performance on previously learned tasks. In addition, this allows for storing multiple specialized models using less disk space. For fine tuning the LLaVA model, we used low-rank adaptation (LoRA) (Hu et al., 2022), which is a method that is particularly relevant within the framework of parameter-efficient fine-tuning (PEFT). This technique allowed us to train only an adapter, which resulted in faster and more resource-efficient training. This adapter layer acts as a bridge between the pretrained LLaVA model and the specific task of triplet extraction from multimodal data that we aimed to optimize it for. By introducing this layer, we enabled the model to better align its image and text representations with the structural requirements of triplet extraction, such as identifying and extracting relevant information in the form of subject–predicate–object triplets.

During inference, a single multimodal prompt was constructed for each target section. This prompt included: (i) the raw section text, (ii) the associated figure or schematic, and (iii) a directive stating "Extract all domain-relevant triplets". This prompt was then processed by our

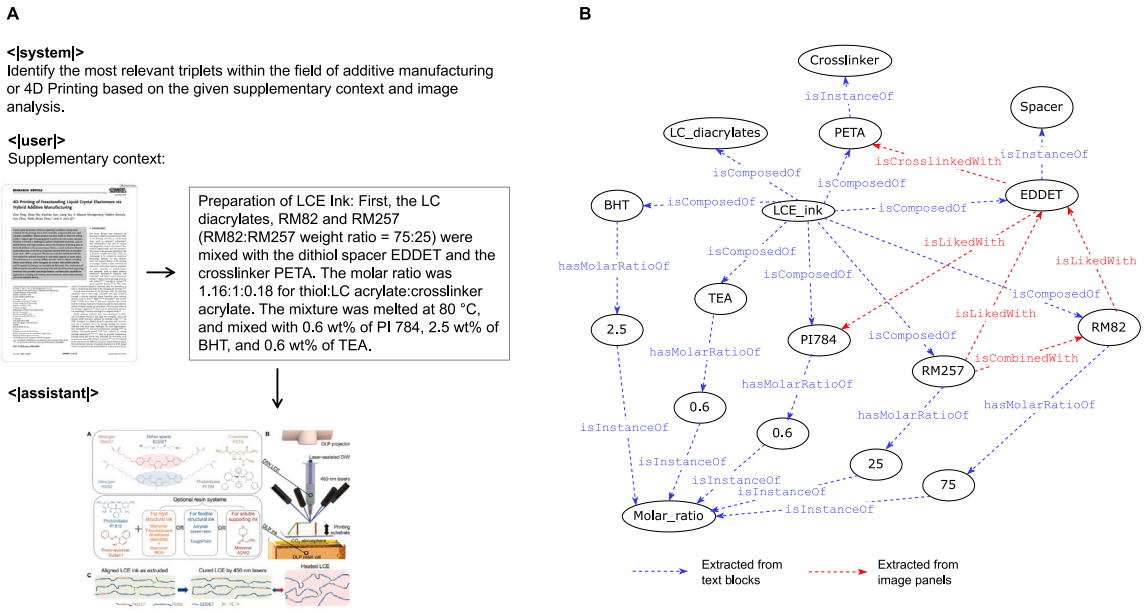


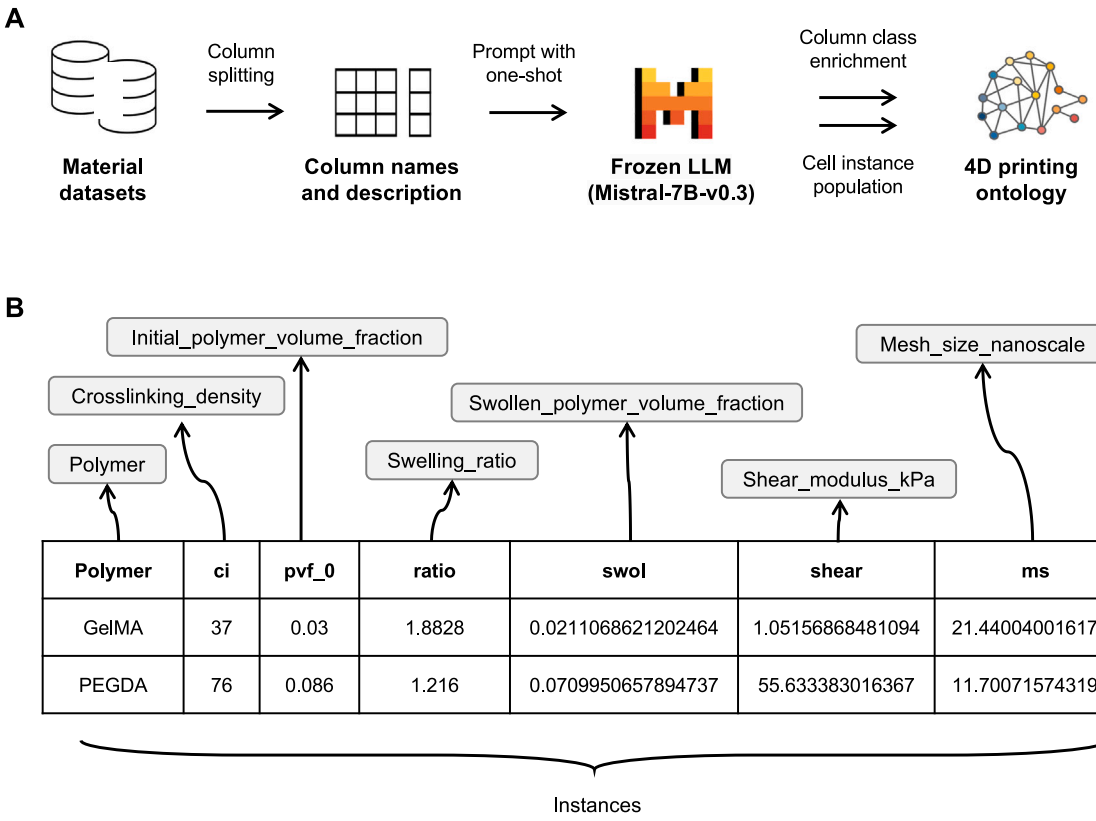
Fig. 7. Example of the inference pipeline: (A) a section is selected from a published article in *Advanced Materials* Journal (Peng et al., 2022), its most relevant image was retrieved, and (B) a corresponding graph of triplets was produced. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

MLLM, which jointly attended to textual tokens and image patches to generate a set of subject–predicate–object assertions. For figures, the model first employed an optical character recognition (OCR) module to detect and encode text regions, and to extract key graphical elements (i.e., shapes, connectors, symbols) as visual tokens. These visual tokens interacted with text embeddings via cross-attention within the multimodal transformer. The text embeddings had been refined through our fine-tuning procedure, therefore allowing for better alignment with domain-specific semantics. This cross-modal mechanism enabled the model to infer high-level semantic relations that are not explicitly stated in the input but emerge from a combination of spatial configurations, textual cues, and prior knowledge encoded in the pretrained weights. For example, in a section from Peng et al.'s paper (Peng et al., 2022), illustrated in Fig. 7A, describing liquid crystal elastomer (LCE) preparation, the spatial proximity and labels of "EDDET" and "PETA" enabled the model to infer an `isCrosslinkedWith` relation, which leverages both the visual structure and domain-specific patterns learned during pretraining. As shown in Fig. 7B, relations grounded purely in the text (e.g., LCE_ink `isComposedOf` TEA) are rendered in blue, whereas those inferred from the figure's spatial layout and graphical elements (e.g., RM257 `isCombinedWith` RM82) appear in red. This integrated multimodal approach thus ensures a reliable extraction of triplets from both explicit textual descriptions and implicit visual patterns.

Our extraction pipeline is intentionally modular, and each component is selected for a distinct, non-overlapping functional role rather than as an interchangeable alternative. LLaVA serves as the multimodal extractor, translating paired textual evidence and figures into ontology-aligned subject–predicate–object triplets. CLIP is used upstream for cross-modal retrieval, aligning each target section with its most semantically relevant figure so that subsequent extraction is grounded in the appropriate visual evidence. A BERT-family encoder provides contextualized text embeddings for similarity-based retrieval and corpus navigation, supporting the evidence selection step that conditions extraction. GPT-4 is used only to construct the fine-tuning supervision set under strict evidence-use constraints; its outputs are treated as candidate annotations and are expert-curated prior to training, ensuring that fine-tuning does not operationalize model generations as ground-truth knowledge. Methodologically, RAG provides evidence grounding by conditioning extraction on retrieved passages and figures, thereby reducing unsupported assertions, while PEFT with LoRA enables resource-efficient adaptation of the large multimodal model under constrained compute by updating only a small subset of parameters. This functional separation clarifies why multiple models appear in the pipeline. They are different in kind – multimodal extraction, cross-modal retrieval, text embedding, and supervision construction – and each is chosen to address a specific bottleneck in the end-to-end knowledge engineering workflow, including grounding, multimodality, scalability, and reproducibility.

2.4. Ontology enrichment from existing datasets

To enhance our 4D printing ontology, we incorporated specific datasets that align with the domain's requirements. Our selection process involved choosing datasets based not only on their relevance but also on their compatibility with the format constraints. In this endeavor, we identified key resources including the Novel Materials Composition and Properties Database (Gnome) (Merchant et al., 2023), sourced from Google DeepMind. This dataset catalogs the compositions and corresponding energies of over 2.2 million new crystalline materials, providing a comprehensive resource for new material discovery and validation. In addition, we utilized the polyOne dataset (Kuenneth and Ramprasad, 2022), which contains 100 million hypothetical polymers, each with 29 predicted properties computed using advanced ML models. The Hydrogel Design dataset (HydrogelDesign, 2023; Richbourg et al., 2021; Richbourg and Peppas, 2023) offers extensive insights into a variety of hydrogel properties. It includes key measurements such as swelling, diffusion, stiffness, and partitioning across different hydrogel formulations. The Smart Materials Database from NC State (Crews et al., 2012) aggregates extensive information on smart materials, including detailed shape memory alloy data such as constant-temperature stress–strain measurements, creep data, and minor loop data. Additionally, the Polymer Property Predictor and Database (PPPDB) from University of Chicago (2023) offers a comprehensive collection of polymer data, enhanced with tools such as the Flory–Huggins Phase Diagram, the Random Phase Approximation Structure Factor, and a Binary Solution Cloud Point Estimator. Materials Project Data (Jain et al., 2013) offers an extensive collection of



Polymer	ci	pvf_0	ratio	swol	shear	ms
GelMA	37	0.03	1.8828	0.0211068621202464	1.05156868481094	21.4400400161711
PEGDA	76	0.086	1.216	0.0709950657894737	55.633383016367	11.7007157431901

Fig. 8. (A) Overview of the dataset pipeline, and (B) illustration of identified classes and instances related to a part of the Hydrogel Design dataset.

computed material properties. The dataset is organized to present a comprehensive overview of each material's characteristics by integrating detailed information on composition, energy stability, electronic features, and structural configurations. Moreover, CoPolDB (Takahashi et al., 2024) focuses on copolymerization by offering two components: a repository of monomers, which serve as the building blocks for copolymers, and a catalog of copolymers generated through polymerization reactions. Although we initially planned to include data from NASA's Shape Memory repository (NASA, 2025), its lack of an API for automated data retrieval limited its integration into our framework.

Our methodology for integrating these datasets into the ontology involves detailed data preparation and mapping. Each dataset was broken down into its constituent columns, which are then described and cataloged with examples provided for clarity. To categorize each column within the 4D printing ontology, we utilized a one-shot learning method (Y. Li et al., 2024; Ucar et al., 2020) with the Mistral-7B-v0.3 LLM model (Jiang et al., 2023). This choice reflects the fact that column-to-class mapping is a text-only task, for which a dedicated high-performing text LLM is sufficient and more appropriate than relying on a heavier multimodal model whose visual capacity would remain unused. The Supplementary Material provides more details about the justification underlying this model choice. We then instantiate each cell in every row as an instance of its corresponding ontology class, as illustrated in Fig. 8 and extract the resource description framework (RDF) object properties that link these instances. For example, in the Hydrogel Design dataset we created instances for classes such as Polymer, SwellingRatio, and ShearModulus_kPa, and generate relations—e.g., Polymer linked via `hasShearModulus` to ShearModulus_kPa, Polymer linked via `hasSwellingRatio` to SwellingRatio, and SwellingRatio linked via `influencesShearModulus` to ShearModulus_kPa. This end-to-end pipeline yields a richly interconnected ontology graph that faithfully captures both the structural typology and the relational semantics of the original data.

In addition to relevance and format compatibility, dataset inclusion was expert-validated with respect to (i) source reliability (provenance, documentation quality, and methodological credibility), and (ii) utility for 4D-printing knowledge modeling (coverage of materials, AM processes, stimuli, and measurable properties). Datasets were excluded when they lacked a programmatic access route (e.g., no API or no feasible automated retrieval), had restrictive licensing terms, were not machine-parseable in practice (e.g., incompatible formats or unstable table structures), provided insufficient documentation, contained incomplete or low-coverage fields, or could not be traced to a reliable, citable source (e.g., no stable repository or peer-reviewed/referenceable record). To further increase the trustworthiness of the ingestion step, we conducted a targeted expert audit of the automatic mapping procedure: we randomly sampled 50 columns across 6 datasets and manually verified the column-to-class assignments proposed by the LLM, which showed high agreement with expert judgment. Finally, for instance enrichment, we treat the numerical and categorical values provided by these curated datasets as the primary data source, since they originate from established repositories or published resources; accordingly, our validation effort focused on semantic typing and alignment (i.e., whether values are assigned to the correct ontology classes and relations) rather than re-measuring or re-estimating the underlying experimental quantities.

2.5. Ontology enrichment from knowledge graph

Furthermore, we harnessed the capabilities of MatKG – that is the most extensive knowledge graph in the field of materials science – to enrich our 4D printing ontology with passive materials. Derived from 5 million articles, MatKG contains over 1 million triplets, which were meticulously extracted using a named entity recognition (NER) model. Termed as MatBERT (Trewartha et al., 2022), this model is a BERT-based system specifically fine-tuned for materials science literature,

enabling precise identification and categorization of entities such as materials, synthesis methods, and properties.

We designed a systematic transformation pipeline to feed our ontology with MatKG's triplets with the following subject-relation-object format. We implemented a targeted relationship mapping strategy to enhance the semantic connections within our ontology. Specifically, we transformed MatKG's domain-specific relation codes into more semantically meaningful predicates: 'CHM-PRO' was converted to '`hasProperty`', where 'CHM' designates materials and 'PRO' refers to properties, while 'CHM-APL' was converted to '`isApplicatedTo`', where 'APL' denotes applications. Additionally, we systematically generated material instance declarations by identifying all chemical entities (CHM) involved in these relationships and explicitly defining them as instances of the Material class through an '`isInstanceOf`' predicate. This end-to-end pipeline thus enabled us to augment our ontology with richly typed data properties and new individual declarations. For example, the original MatKG triplets [PNIPAM, CHM-PRO, LCST] and [PNIPAM, CHM-APL, Drug Delivery] – where PNIPAM and LCST designate Poly(N-isopropylacrylamide) and Lower Critical Solution Temperature, respectively – were converted into the following semantically enriched relationships: [PNIPAM, `isInstanceOf`, Active_Material], [PNIPAM, `hasProperty`, LCST], and [PNIPAM, `isApplicatedTo`, Drug delivery].

2.6. Preprocessing and construction of the ontology

The results obtained from this framework were cleaned through a rigorous process that ensures only high-quality triplets are retained. Specifically, the framework evaluates the following aspects:

- **Domain relevance:** Each triplet's subject and object are transformed into contextualized embeddings using BERT and compared against embeddings derived from a curated list of domain-specific keywords. For each triplet element, the framework computes cosine similarity scores against all domain keywords and retains the maximum similarity value as the relevance indicator. The final domain relevance assessment combines both subject and object relevance scores in the overall evaluation function. This process ensures that the data is deeply aligned with the target field.
- **Semantic coherence:** The framework implements a comprehensive evaluation strategy to assess semantic meaningfulness. It computes direct BERT-based cosine similarity between subject and object embeddings to measure their semantic relatedness. The final coherence score integrates both the relation validity and subject–object similarity components. Additionally, predicate coherence is evaluated through template-based assessment, where the framework compares BERT embeddings of complete triplet phrases against baseline phrases and relationship templates to ensure predicate appropriateness within the semantic context.
- **Structural validity:** The framework checks the syntactic correctness of each triplet by verifying that all elements (subject, predicate, and object) are present, of sufficient length, and follow expected formatting standards. This validation ensures data reliability for downstream applications.
- **Redundancy elimination:** Duplicate or highly similar triplets are identified through a two-stage process. First, exact duplicates are removed through string matching of subject–predicate–object combinations. Second, semantic duplicates are detected by computing BERT-based cosine similarity between triplet embeddings, where triplets exceeding a similarity threshold are flagged as redundant. This ensures that the final dataset is concise and free from both literal and semantic redundancy.

Together, these validation steps contribute to a robust and high-fidelity cleaning process that prepares the data for subsequent ontology construction and analysis. In addition to these quality control measures, the ontology construction phase integrated several advanced techniques to further enhance the ontology. First, entity names are normalized and cleaned to create valid uniform resource identifier fragments, thereby ensuring semantic consistency across the ontology. This preprocessing step effectively mitigates errors arising from formatting discrepancies or lexical variations. Furthermore, the framework incorporates a BERT-based similarity analysis that compares new class labels with those already present in the ontology. This mechanism dynamically identifies semantically similar classes and, when a sufficient similarity threshold is met, establishes subclass relationships. In doing so, the ontology consolidates redundant entities and organizes them hierarchically in a manner that mirrors the underlying domain structure. The pipeline also incorporates robust logging and error handling throughout each processing stage. Detailed logs capture any discrepancies or skipped entries, which facilitates iterative refinement and ensures full traceability of the data transformation process. Moreover, special attention has been given to maintaining the homogeneity of the complete ontology by enforcing uniform naming conventions and consistent semantic representations across all entities. This ensures that the entire knowledge base exhibits a high degree of internal consistency, which is critical for efficient reasoning and data integration.

3. Results

3.1. Final enriched 4D printing ontology

Our framework initiates the ontology enrichment process with an initial 4D printing ontology, which comprises only 170 classes, 9 instances, 48 object properties, and 13 data properties. Through the successive integration of heterogeneous data sources and advanced validation techniques, the framework has dramatically enriched and populated the ontology. In the first phase, the system processed a corpus of scientific articles by extracting triplets that describe various domain-specific relationships. In total, approximately 130,000 triplets were initially generated from articles, of which only about 28,000 were retained after applying the full consistency filtering pipeline, including semantic similarity checks, predicate template matching, and redundancy elimination. This stage resulted in the identification of 5706 classes, 16,651 instances, 1331 object properties, 4390 data properties, and the establishment of 7913 subclass relationships. In addition to this large-scale quantitative enrichment, we conducted a systematic evaluation of the triplet-extraction model itself, as detailed in the Supplementary Materials (See Figure S1). The consideration of MatKG further augmented the ontology by processing additional instance-of relationships. It was responsible for incorporating 6629 new instances and two additional data properties with 445,370 relations. This considerable increase reflects the framework's ability to integrate detailed instance-level data from supplementary sources, thereby enhancing the granularity and applicability of the ontology. A further enrichment occurred through the automated ingestion of multiple datasets from an external directory. This step contributed 144 additional classes, 12,540,671 instances, 26 object properties, and 113 subclass relationships (see Fig. 9). By parsing and merging these large-scale datasets, the framework ensured a comprehensive and diverse coverage of the domain knowledge, while maintaining structural validity and eliminating redundancy.

After synthesizing the contributions from the scientific literature, the MatKG module, and additional datasets, the final ontology exhibits 5849 classes, 12,563,951 instances, 1357 object properties, 4392 data properties, and 8196 subclass relationships. This substantial ontology expansion demonstrates the efficacy of our multi-stage enrichment process, as illustrated in Fig. 10 and **Movie S1**, which highlights the

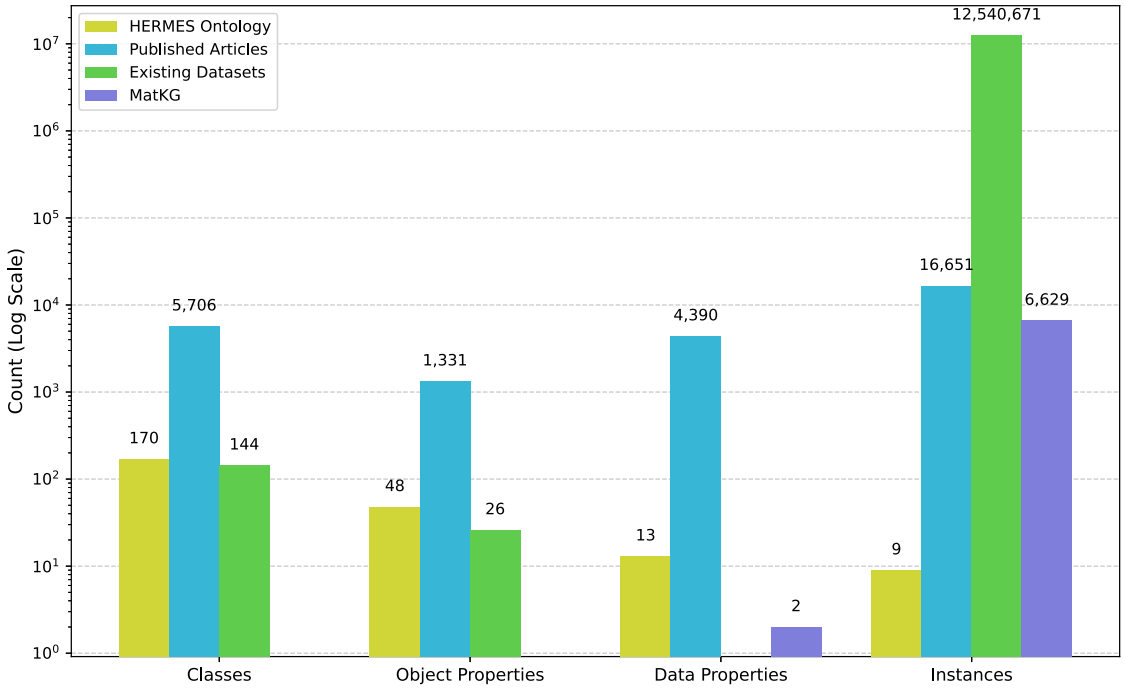


Fig. 9. Comparison of ontology components between the baseline HERMES ontology and its extended counterparts derived from published articles processing, dataset parsing, and integration with MatKG triplets. The values demonstrate the significant scalability of automated multimodal extraction and knowledge graph integration.

material, design-and-engineering, and AM views, along with a subgraph centered on the stimuli-sensitive Hydrogel class (Fig. 11).

In summary, the integration of multiple data sources, coupled with advanced NLP and robust validation measures, has culminated in a high-fidelity, richly structured ontology. This framework is fully domain-agnostic—while it was demonstrated on 4D printing, it can just as easily be applied to any other field. The resulting ontology not only represents a substantial expansion in scale and detail compared to its initial state but also provides a solid foundation for downstream applications such as knowledge-based reasoning, data integration, and semantic information retrieval across complex scientific and technical domains.

3.2. Query–answer examples for smart materials and structures

The enriched ontology enables the formulation and resolution of design-oriented queries in 4D printing, aligning with recent industrial question-answering systems that support technical knowledge access through natural language interfaces (Ruiz et al., 2023). To demonstrate its relevance, we encoded a material selection need as a semantic query vector in the ontology embedding space and retrieved the most relevant materials, experiments, and architectures by ranking their embeddings using cosine similarity. In this illustrative case, the design intent was: "Identify a soft, biocompatible material that is chemically compatible with LCE and printable via direct ink writing (DIW)".

The highest-ranked neighborhood returned by cosine similarity was dominated by silicone-based elastomers. This ranking was supported by convergent evidence from the literature: (i) DIW processable silicones, including polydimethylsiloxane (PDMS) formulations, (ii) their established biocompatibility in biomedical and microdevice applications, and (iii) explicit LCE–silicone co-fabrication reports indicating chemical/process compatibility at the interface.

As the top-ranked result, the ontology retrieved PDMS as a soft silicone elastomer that can be processed by DIW using embedded/freeform strategies, where uncured PDMS inks were extruded into a hydrophilic support bath and subsequently cured into stable 3D features (Hinton et al., 2015; Li and Li, 2022). PDMS is also widely documented as biocompatible and chemically inert, which explains its routine use in soft implantable systems (McDonald and Whitesides, 2002). Crucially for LCE compatibility, PDMS has been used as the passive silicone layer in bilayer systems integrating cholesteric or nematic LCE actuators, demonstrating that silicone-LCE interfaces can be formed and operated without reported interfacial chemical inhibition. Jiang et al. (2024) and M. Li et al. (2024) Representative extracted triples include [PDMS, isPrintedUsing, DIW], [PDMS, is, Biocompatible], and [LCE actuator, isIntegratedWith, PDMS layer], confirming PDMS as a DIW-printable, biocompatible material compatible with LCE systems.

Beyond PDMS, the ontology identified DIW-printable soft silicone elastomer inks formulated from platinum-cured or room-temperature-vulcanizing (RTV) systems. These materials are reported as DIW-processable into highly compliant parts for soft devices, and they share the same chemically inert polysiloxane backbone class as PDMS, which is the silicone family already demonstrated in LCE-integrated constructs (Hinton et al., 2015). Extracted triples such as [Soft silicone elastomer, isPrintedUsing, DIW], [Soft silicone elastomer, hasBackbone, Polysiloxane], and [Polysiloxane, isCompatibleWith, LCE layer] further support DIW-printable silicones as the closest validated compatibility class for co-printing with LCE actuators.

We then evaluated a second design need: "Determine the material distribution of LCE and PDMS to ensure actuation and biocompatibility of the structure". The ontology returned an embedded composite architecture, in which discrete LCE particles are dispersed within a continuous PDMS elastomer matrix. This configuration corresponds to reported polymer-dispersed LCE composites, where LCE particles provide localized anisotropic actuation and the PDMS matrix serves as a compliant host that transmits deformation while avoiding mechanical clamping of the LCE domains. When the PDMS matrix is sufficiently soft, the LCE inclusions can undergo reversible actuation, yielding macroscopic shape change of the composite. This architecture simultaneously benefits from the established biomedical compatibility of PDMS in soft device applications. Representative triples supporting this match include [LCE particles, isDispersedIn, PDMS matrix], [LCE

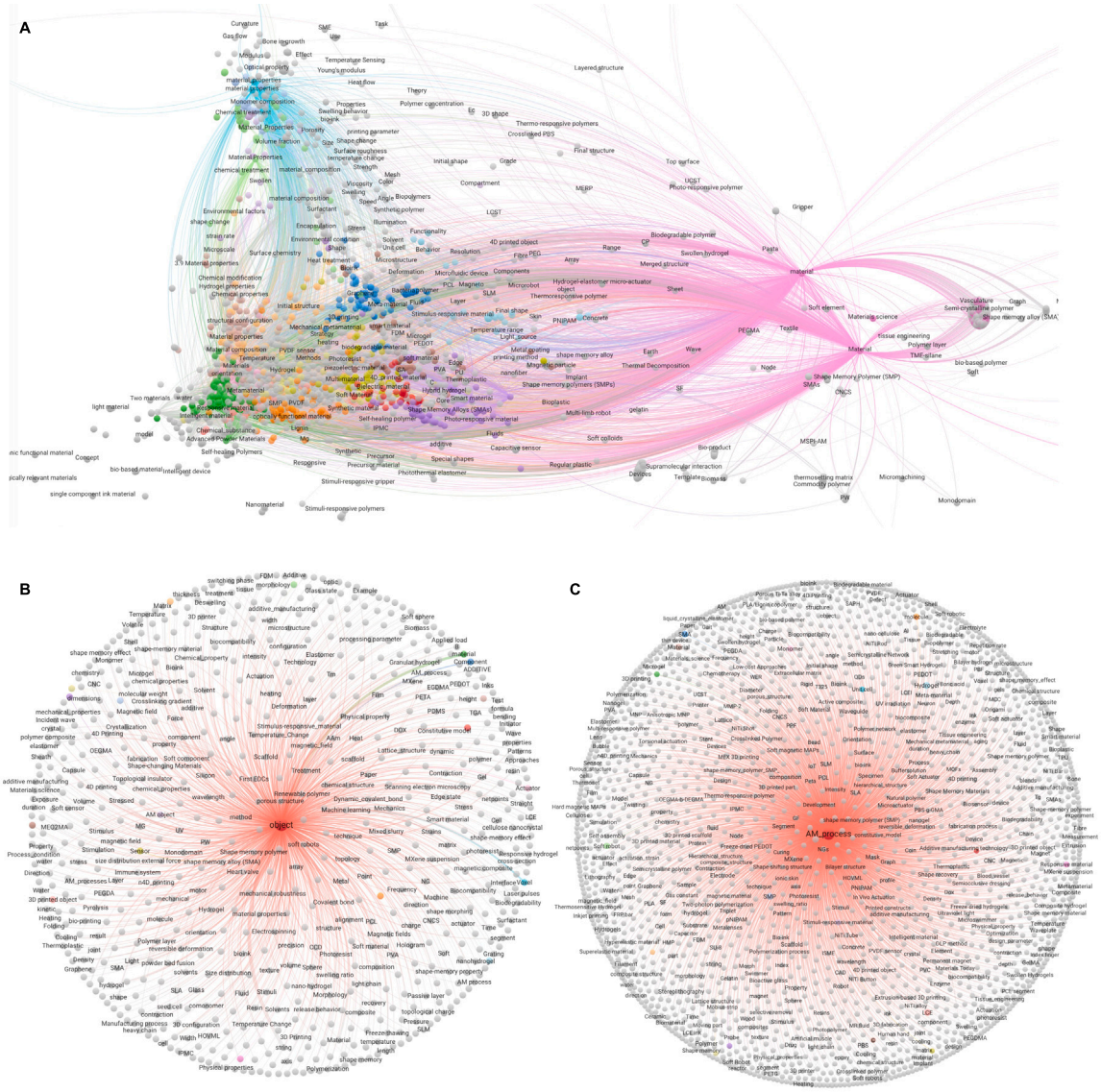


Fig. 10. Final enriched 4D printing ontology illustrated through its (A) material view, (B) design and engineering view, and (C) AM view.

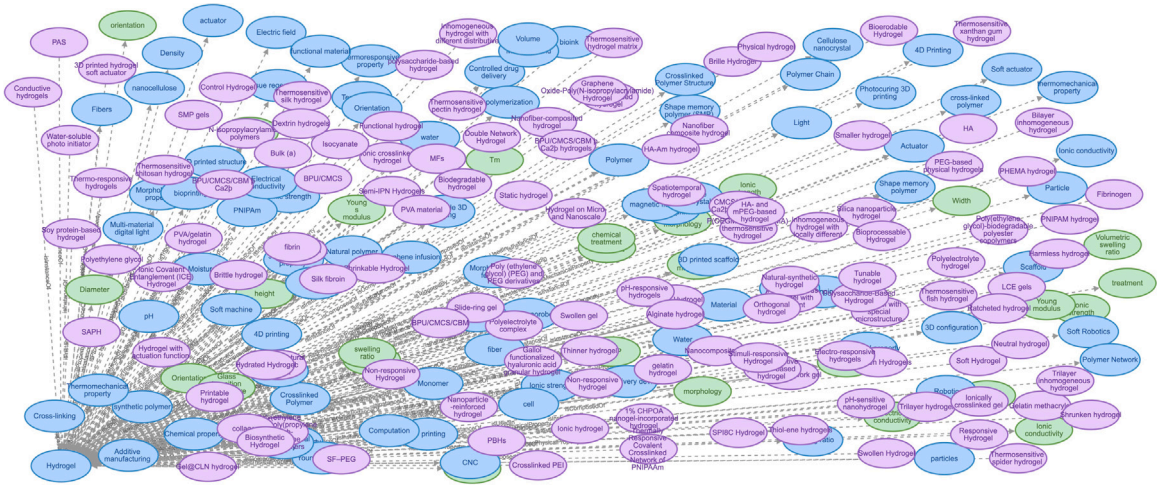


Fig. 11. Final enriched 4D printing ontology through the lens of the stimuli-sensitive Hydrogel class. To ensure clarity, only a limited number of classes, object properties, and instances are displayed.

composite, exhibits, Reversible actuation], and [PDMS, is, Biocompatible], all grounded in reported studies on LCE–PDMS composites and PDMS-based biomedical systems (Bobnar et al., 2023; McDonald and Whitesides, 2002).

The enriched ontology demonstrated strong potential as a design-support system for 4D printing by enabling high-level, intent-driven queries to be translated into relevant recommendations. By embedding design intents into a semantic vector space and retrieving knowledge via similarity ranking, the ontology accurately identified DIW-printable, biocompatible silicone elastomers – particularly PDMS – as optimal matches for a active, LCE-compatible material, with results supported by literature-derived triples and experimental reports. It further inferred an appropriate LCE–PDMS composite architecture, retrieving evidence for embedded LCE domains within a compliant PDMS matrix that preserves actuation while ensuring biocompatibility. These examples illustrate the ontology's ability to integrate heterogeneous knowledge, infer mechanistically coherent solutions, and guide material and structural design decisions, highlighting its promise as a generalizable and evidence-backed tool for accelerating discovery in smart material systems.

To move beyond purely illustrative query–answer demonstrations, we compared the two competency questions against (i) the baseline HERMES ontology and (ii) our enriched ontology along three technically grounded dimensions: representational completeness (whether the ontology contains the required entities and relations to express the intent), instance-level operationality (whether the query can be resolved to concrete candidates rather than remaining at the schema level), and evidential traceability (whether returned assertions are supported by explicitly extracted triples anchored in the source corpus). For the first question, HERMES supports only a schema-level interpretation: it represents DIW as a manufacturing concept (e.g., `Direct_ink_writing` under `Material_extrusion`) and provides generic modeling patterns such as `Object` → `hasMaterial` → `Material`, but it neither populates candidate material instances (e.g., PDMS) nor represents the constraint vocabulary needed to formalize the design intent (e.g., biocompatibility and LCE–silicone chemical/process compatibility). As a result, the competency question cannot be instantiated into a design-resolvable query and yields, at best, high-level conceptual matches. For the second question, HERMES contains structural and functional primitives that could, in principle, encode transformable systems (e.g., voxel-based constructs such as `Voxel` arranged in a `Voxel_grid`, and patterns such as `Stimulus` → `enables` → `Transformation_function`); however, it does not represent compositional micro-architectures (e.g., "particles dispersed in a matrix") and again lacks the instance-level entities and typed relations required to connect materials, architectures, and performance-relevant properties, preventing resolution beyond generic statements about objects and transformations. In contrast, the enriched ontology resolves both questions at the instance and evidence levels by returning concrete candidates (e.g., PDMS and related DIW-processable silicone elastomers) and an embedded LCE–PDMS composite architecture, each supported by explicit, literature-derived triples. This enrichment strengthens the ontology's role as an AI-assisted design substrate in three complementary ways. From a reasoning perspective, this approach increases semantic controllability by encoding key design requirements as well-typed, cross-linked assertions (material-process-property-architecture), enabling precise constraint-oriented querying (e.g., "DIW-printable" ∧ "soft/compliant" ∧ "biocompatible" ∧ "compatible with LCE interfaces") and reducing reliance on implicit narrative interpretation. In terms of decision support, it enables structured candidate narrowing by linking materials to processes, properties, and integration strategies, thereby supporting reproducible screening of material-process-architecture alternatives and providing evidence-backed rationales through the underlying extracted triples. From a knowledge discovery standpoint, it supports structural discovery from the aggregated corpus by (i) introducing new explicit relations between previously weakly connected entities, (ii) surfacing recurrent association patterns as graph motifs (e.g., bilayer and embedded-composite strategies coupling passive silicone matrices with active LCE phases), and (iii) revealing coverage gaps, i.e., plausible material–process-architecture links that are sparsely supported, thereby guiding follow-up experimentation or simulation and closing the loop between data-driven extraction, symbolic structuring, and design-relevant hypothesis generation.

4. Discussion

This substantial increase in conceptual scope provides a robust foundation for the precise representation of components and relationships inherent to transformable systems. Leveraging the breadth and depth of the modeled concepts, one can now intricately characterize both the functional and geometric configurations these systems may assume. Such enhanced granularity creates a fertile substrate for the development of more realistic simulation scenarios and the formulation of optimization criteria tailored to the specific constraints of each application. Ultimately, this enriched ontology could serve as a guiding framework to inform and streamline subsequent design phases. This originality aligns with recent perspectives (Kayusi et al., 2025), which highlight how novel frameworks integrating large-scale AI models with domain-specific tasks can expand the boundaries of knowledge representation.

The 4D printing ontology provides a formalized vocabulary and structure for the key 4D printing concepts, i.e, stimuli, smart materials, transformation functions, and AM techniques, into a coherent schema that supports both human understanding and machine processing (Bougzime et al., 2025). By adhering to OWL (McGuinness et al., 2004) and Industry Ontology Foundry (IOF) guidelines, the ontology ensures that terms are precisely defined, relationships are explicitly modeled, and extensions remain compatible with related ontologies in materials science and manufacturing. This semantic rigor facilitates robust data exchange and tool integration across computer-aided design (CAD) systems, simulation packages, knowledge management platforms, and knowledge-driven digital twin environments (Su et al., 2024).

Ontologies serve as the backbone for reasoning engines that can infer new knowledge, check consistency, and support design trade-offs. In the context of 4D printing, rules built on top of the ontology can evaluate which stimuli-material arrangements yield desired transformations, recommend printing parameters, and flag incompatible design options. Such rule-based reasoning complements data-driven predictions from neural models, providing explainable decision paths that are critical for certification and compliance, and knowledge-driven decision support in engineering contexts (Su et al., 2024; Oyekan et al., 2025).

4D printing lies at the intersection of materials science, mechanical design, computational mechanics, chemistry, and computer science, each with its own lexicon and methodologies (Dimassi et al., 2023). The ontology enables these diverse experts – product architects, material scientists, AM process planners – to collaborate within a unified semantic framework, reducing misinterpretations and ensuring that design intents are faithfully communicated, which is consistent with recent knowledge-augmented approaches for industrial human–machine collaboration (Oyekan et al., 2025). By mapping domain-specific data (e.g., simulation outputs, experimental results) to shared ontology classes and properties, it ensures semantic alignment and coherent knowledge integration across multidisciplinary workflows.

When coupled with neuro-symbolic AI systems, the 4D printing ontology becomes a dynamic asset: NSAI can ingest unstructured text and visuals, map extracted entities and relations to ontology terms, and trigger reasoning processes for immediate insights. This tight integration supports continuous ontology evolution through new materials,

stimuli, or manufacturing strategies discovered in the literature can be auto-added and validated against existing constraints. In practice, this accelerates the iteration cycle for 4D printed designs, enabling rapid prototyping informed by both data-driven predictions and symbolic rules, and ensuring the design pipeline remains synchronized with the cutting edge of materials research and data-driven design practices (Lee and Ahmed-Kristensen, 2025).

Despite these encouraging outcomes, our approach is not without its critiques. The framework currently relies heavily on the quality and representativeness of the input data, and its scalability remains a challenge, particularly regarding processing time and resource demands. In addition, potential errors in the underlying datasets can propagate into the ontology. In this study, the datasets we used were validated by domain experts, which ensured a high level of reliability. However, as the framework scales to millions of instances, maintaining such expert validation becomes increasingly challenging, underlining the need for automated yet robust validation mechanisms in future work. Moreover, while the present study focused on 4D printing, the proposed methodology is designed to be transferable: the same combination of triplet extraction, consistency filtering, and ontology enrichment could be applied to other domains, provided appropriate corpora and expert validation are available. However, we recognize that this claim of domain-agnosticism should be tempered until validated experimentally in additional fields. Our experiments were conducted under constrained hardware resources (e.g., CPU/GPU specifications). Within these constraints, we relied exclusively on open-source models to ensure transparency and reproducibility. Access to more powerful hardware would enable the use of more advanced LLMs, which we expect to improve both the precision and efficiency of the ontology-enrichment process. Such advancements would further enhance the system's ability to extract, reason, and update domain-specific knowledge in real time. In summary, while our study marks a significant step towards more intelligent, adaptable, and scalable ontology-driven systems in the rapid evolving 4D printing, future work should focus on addressing the current limitations by incorporating broader datasets and leveraging next-generation language models, ultimately fostering more robust and comprehensive knowledge management solutions in emerging fields.

A central methodological consideration in applying probabilistic LLMs to OWL ontology enrichment is the epistemic risk introduced when stochastic generations are translated into symbolic commitments. This risk is compounded by corpus-level limitations: because the distribution of topics in the scientific literature is uneven, enrichment may over-represent well-studied domains while under-representing emerging or peripheral subfields. Moreover, there is a risk of overfitting the ontology to currently available literature, which may reduce its adaptability as new concepts and paradigms emerge. Addressing these limitations requires designing the ontology with mechanisms that preserve flexibility and scalability. This design tension echoes principles emphasized in organic architecture, namely maintaining structural rigor while preserving adaptability to changing contexts (Abdulkareem, 2024). In this setting, four threats are particularly salient. First, semantic drift refers to progressive meaning shifts that can arise during iterative extraction, canonicalization, and re-alignment. Second, latent bias may be inherited from pretraining distributions and from uneven coverage in the retrieved corpus. Third, noisy co-occurrence patterns in scientific text can be reified into seemingly authoritative ontological axioms even when the underlying evidence is weak or highly context-dependent. Fourth, context loss can lead to over-generalization when conditional claims are expressed without the qualifiers needed to delimit their validity. Many statements in 4D printing are valid only under specific experimental conditions. Therefore, relations extracted from the literature, such as "DIW-printable" or "material compatibility", should be interpreted as context-dependent and should not be promoted to general ontological assertions unless the relevant qualifiers are captured, including the material formulation, the processing parameters, and the stimulus conditions. For this reason, the pipeline treats LLM outputs as candidate assertions rather than direct OWL commitments. Before any assertion is committed to the ontology, it is subjected to the quality gates described earlier, which jointly enforce domain alignment, semantic plausibility of the relation, basic structural well-formedness, and redundancy control. This separation between probabilistic proposal and symbolic acceptance mitigates self-reinforcing feedback loops in which model-generated relations could otherwise be re-ingested as self-justifying axioms, and it helps maintain an auditable epistemic status suitable for OWL-based knowledge engineering. Nevertheless, residual model- and tool-induced errors may persist at scale, motivating future work on uncertainty-aware representations, stronger provenance formalization, and scalable expert-in-the-loop verification.

We acknowledge that, even with expert validation on curated subsets of articles and extracted outputs, human-in-the-loop verification cannot realistically scale to the full volume of documents, figures, and instances processed by our pipeline. Consequently, residual tool- and model-induced errors may persist and should be treated as threats to validity rather than assumed away. We further recognize that the software toolchain used for corpus construction and parsing can introduce systematic biases. ResearchRabbit may skew literature coverage toward highly connected regions of citation graphs seeded by the chosen keywords (topic-centrality bias), potentially under-representing emerging or peripheral subfields in smart materials and 4D printing. LLM Sherpa may introduce segmentation and chunking artifacts when parsing scientific PDFs (boundary bias), which can affect retrieval granularity and propagate to downstream extraction. Aspose-based conversion and figure extraction may also mis-handle complex layouts (e.g., multi-panel figures, caption alignment, or embedded tables), producing layout-dependent noise and occasional mis-associations between evidence and extracted claims. To mitigate these risks, we apply deduplication and controlled keyword expansion during corpus construction, perform sampling-based manual inspection of parsed outputs, and cross-check text–figure provenance during retrieval. In addition, we rely on downstream ontological validation (e.g., domain/range typing, redundancy filtering, and consistency-oriented reasoning checks) to reduce the likelihood that structurally inconsistent, weakly typed, or insufficiently grounded assertions are committed to the knowledge base. Nevertheless, these safeguards do not eliminate all errors, and future work should prioritize scalable verification strategies and uncertainty-aware representation of extracted knowledge.

Recent advancements in language model capabilities, including their use as digital intelligent assistants in manufacturing, suggest promising new directions (Colabianchi et al., 2024). One significant improvement could involve implementing a multi-agent LLM framework for automated, multimodal knowledge extraction and integration from unstructured scientific texts and visuals. A multi-agent LLM system could significantly enhance our ontology enrichment pipeline by deploying specialized agents that work in concert, each handling distinct aspects of knowledge ingestion—from text segmentation and entity recognition to figure interpretation, table parsing, relationship extraction, relation-type classification, schema alignment, and conflict resolution (Trirat et al., 2024; Lu and Wang, 2025). Such an architecture enables fine-grained processing of heterogeneous content, dramatically broadening the scope of automated ontology construction (Babaei Giglou et al., 2023).

First, the division of labor among specialized agents enables more nuanced analysis of domain-specific terminology and modalities (Katzer et al., 2025). Dedicated text agents can be fine-tuned for extracting materials entities and technical nomenclature, while vision–language agents process figures (e.g., microstructure images, X-ray diffraction patterns) to recognize visual evidence of material phases and morphologies (Buehler, 2024; Polak and Morgan, 2025), and table agents convert tabular data into structured triplets (Khalighinejad et al., 2025). Relation classification agents specialize in categorizing extracted relations into classes, object properties, data properties, or instances,

ensuring accurate semantic typing per OWL standards (Ciatto et al., 2025).

Second, the integration of cross-agent verification mechanisms would substantially increase the reliability of extracted knowledge (Vsevolodovna and Monti, 2025). By implementing a system where schema alignment agents validate candidate entities and relations against ontological constraints, vision agents perform cross-modal consistency checks between text descriptions and corresponding figures, and conflict resolution agents evaluate logical coherence among new and existing triplets, thus minimizing spurious connections in the ontology. Additionally, a cohesion verification agent continuously monitors the ontology's structural integrity – detecting hierarchy cycles, disjointness violations, and domain/range inconsistencies – while a triplet validation agent flags semantically incomplete or logically incoherent triples for further review (Amini et al., 2024).

Third, this multimodal approach could dramatically improve scalability. The current exponential growth in materials science literature – with thousands of new papers published monthly – demands automated systems capable of processing information at scale while maintaining precision (Katzer et al., 2025). A multi-agent framework could adaptively distribute computational resources based on document complexity and domain relevance (Yang et al., 2024). Furthermore, such a system could implement domain-adaptive prompting strategies, allowing it to navigate the specialized terminology of different materials science subdomains while preserving semantic accuracy across the entire ontology (Saeedizade and Blomqvist, 2024).

Future work will focus on (i) designing specialized agent architectures that integrate vision encoders and domain-specific prompt templates for materials science modalities, (ii) implementing advanced verification heuristics that leverage both linguistic and visual ontological rules, and (iii) developing evaluation metrics for multimodal triplet extraction that reflect the unique challenges of materials knowledge representation, and (iv) creating dedicated relation classification agents for precise typing and specialized validation agents for ontology cohesion and triplet integrity. By embracing multimodal multi-agent systems, we can move toward adaptive ontologies that evolve seamlessly with the scientific literature, providing researchers with powerful tools for accelerated materials discovery and development.

5. Conclusion

In this work, we presented an innovative framework for ontology learning in the field of 4D printing, integrating MLLMs and RAG to overcome the limitations of traditional ontological systems. Our approach, successfully combines the formal rigor of structured knowledge representation with the adaptive and contextual capabilities of advanced language models, which systematically captures heterogeneous information from scientific literature, experimental databases, CSV files, and extensive knowledge graphs like MatKG. Experimental results demonstrate that our methodology significantly expanded an initial, rather limited ontology – starting from 170 classes and a few instances – to a comprehensive structure encompassing over 5800 classes and more than 12.5 million instances.

The enriched and populated ontology promotes reuse and integration with other domain ontologies, therefore reducing silos and fostering cross-domain interoperability (Dimassi et al., 2021a). Importantly, this structured representation lays the groundwork for automated reasoning procedures, enabling qualitative decision support in material and process selection, design recommendations, and alignment with emerging standards. When embedded within a neuro-symbolic AI framework, the ontology can be updated in real-time and reasoned over alongside neural models, bridging symbolic and neural approaches for a context-aware design strategy (Bougzime et al., 2025).

CRediT authorship contribution statement

Oualid Bougzime: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Christophe Cruz:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Kun Zhou:** Writing – review & editing, Validation, Methodology. **H. Jerry Qi:** Writing – review & editing, Validation, Methodology, Conceptualization. **Frédéric Demoly:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compind.2026.104522.

Data availability

Data will be made available on request.

References

Abdulkareem, K.R., 2024. Essence and proposals of organic architecture. Mesopotamian J. Civ. Eng. 2024 (3), 29–34. http://dx.doi.org/10.58496/MJCE/2024/005, URL: https://mesopotamian.press/journals/index.php/MJCE/article/view/566.

Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al., 2023. GPT-4 technical report. arXiv preprint arXiv:2303.08774.

Alam, M., Birkholz, H., Dessì, D., Eberl, C., Fliegl, H., Gumbsch, P., von Hartrott, P., Mädler, L., Niebel, M., Sack, H., Thomas, A., 2021. Ontology modelling for materials science experiments. In: Joint Proceedings of the Semantics Co-Located Events: Poster & Demo Track and Workshop on Ontology-Driven Conceptual Modelling of Digital Twins (Co-Located with SEMANTiCS 2021). In: CEUR Workshop Proceedings, vol. 2941, CEUR-WS.org, pp. 1–12, URL: http://ceur-ws.org/Vol-2941/paper11.pdf.

Amini, R., Norouzi, S.S., Hitzler, P., Amini, R., 2024. Towards complex ontology alignment using large language models. In: International Knowledge Graph and Semantic Web Conference. Springer, pp. 17–31.

Arp, R., Smith, B., Spear, A.D., 2015. Building Ontologies with Basic Formal Ontology. The MIT Press.

Aspose, 2024. Aspose.PDF for Python via .NET. https://pypi.org/project/aspose-pdf/.

Babaei Giglou, H., D'Souza, J., Auer, S., 2023. LLMs4OL: Large language models for ontology learning. In: The Semantic Web – ISWC 2023. In: Lecture Notes in Computer Science, vol. 14265, Springer, pp. 408–427. http://dx.doi.org/10.1007/978-3-031-47243-6_25.

Bobnar, M., Derets, N., Umerova, S., Domenici, V., Novak, N., Lavrič, M., Cordoyiannis, G., Zalar, B., Rešetič, A., 2023. Polymer-dispersed liquid crystal elastomers as moldable shape-programmable materials. Nat. Commun. 14, 764. http://dx.doi.org/10.1038/s41467-023-36426-y.

Bougzime, O., Cruz, C., André, J.-C., Zhou, K., Qi, H.J., Demoly, F., 2025. Neuro-symbolic artificial intelligence in accelerated design for 4D printing: Status, challenges, and perspectives. Mater. Des. 113737. http://dx.doi.org/10.1016/j.matdes.2025.113737.

Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al., 2020. Language models are few-shot learners. In: Advances in Neural Information Processing Systems. vol. 33, pp. 1877–1901.

Buehler, M.J., 2024. Cephalo: Multi-modal vision-language models for bio-inspired materials analysis and design. Adv. Funct. Mater. http://dx.doi.org/10.1002/adfm.202409531, Published version; see also arXiv:2405.19076.

Cheung, K., Drennan, J., Hunter, J., 2008. Towards an ontology for data-driven discovery of new materials.. In: AAAI Spring Symposium: Semantic Scientific Knowledge Integration. pp. 9–14.

Ciatto, G., Agiollo, A., Magnini, M., Omicini, A., 2025. Large language models as oracles for instantiating ontologies with domain-specific knowledge. Knowl.-Based Syst. 310, 112940.

Cimiano, P., Völker, J., 2005. Text2onto: A framework for ontology learning and data-driven change discovery. In: International Conference on Application of Natural Language To Information Systems. Springer, pp. 227–238.

Colabianchi, S., Costantino, F., Sabetta, N., 2024. Assessment of a large language model based digital intelligent assistant in assembly manufacturing. Comput. Ind. 162, 104129. http://dx.doi.org/10.1016/j.compind.2024.104129.

Crews, J.H., Smith, R.C., Pender, K.M., Hannen, J.C., Buckner, G.D., 2012. Data-driven techniques to estimate parameters in the homogenized energy model for shape memory alloys. J. Intell. Mater. Syst. Struct. 23 (17), 1897–1920.

Curtarolo, S., Setyawan, W., Wang, S., Xue, J., Yang, K., Taylor, R.H., Nelson, L.J., Hart, G.L., Sanvito, S., Buongiorno-Nardelli, M., et al., 2012. AFLOWLIB.ORG: A distributed materials properties repository from high-throughput ab initio calculations. Comput. Mater. Sci. 58, 227–235.

Demoly, F., André, J.-C., 2021a. Is order creation through disorder in additive manufacturing possible? In: Pham, D. (Ed.), Cogent Eng. 8 (1), 1889110. http://dx.doi.org/10.1080/23311916.2021.1889110.

Demoly, F., André, J.-C., 2021b. Research strategy in 4D printing: Disruptive vs incremental? J. Integr. Des. Process. Sci. 24 (2), 53–73. http://dx.doi.org/10.3233/JID200020.

Demoly, F., André, J.-C., 2022. 4D Printing, Volume 1: Between Disruptive Research and Industrial Applications. John Wiley & Sons.

Demoly, F., André, J.-C., 2024. 4D printing: Bridging the gap between fundamental research and real-world applications. Appl. Sci. 14 (13), 5669. http://dx.doi.org/10.3390/app14135669.

Demoly, F., Dunn, M.L., Wood, K.L., Qi, H.J., André, J.-C., 2021. The status, barriers, challenges, and future in design for 4D printing. Mater. Des. 212, 110193.

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of NAACL-HLT. Vol. 1, Minneapolis, Minnesota, p. 2.

Dimassi, S., Demoly, F., Belkebir, H., Cruz, C., Kim, K.-Y., Gomes, S., Qi, H.J., André, J.-C., 2023. A knowledge recommendation approach in design for multi-material 4D printing based on semantic similarity vector space model and case-based reasoning. Comput. Ind. 145, 103824.

Dimassi, S., Demoly, F., Cruz, C., Gomes, S., 2021a. From dispersed knowledge to ontology: A proposal for formalizing and integrating 4D printing in design. In: IFIP International Conference on Product Lifecycle Management. Springer, pp. 80–95.

Dimassi, S., Demoly, F., Cruz, C., Qi, H.J., Kim, K.-Y., André, J.-C., Gomes, S., 2021b. An ontology-based framework to formalize and represent 4D printing knowledge in design. Comput. Ind. 126, 103374.

Dinar, M., Rosen, D.W., 2017. A design for additive manufacturing ontology. J. Comput. Inf. Sci. Eng. 17 (2), 021013.

Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.-M., Chen, W., et al., 2023. Parameter-efficient fine-tuning of large-scale pre-trained language models. Nat. Mach. Intell. 5 (3), 220–235.

Doumanas, D., Soularidis, A., Spiliotopoulos, D., Vassilakis, C., Kotis, K., 2025. Fine-tuning large language models for ontology engineering: A comparative analysis of GPT-4 and Mistral. Appl. Sci. 15 (4), 2146.

Draxl, C., Scheffler, M., 2018. NOMAD: The FAIR concept for big data-driven materials science. Mrs Bull. 43 (9), 676–682.

Fu, Z., Yang, H., So, A.M.-C., Lam, W., Bing, L., Collier, N., 2023. On the effectiveness of parameter-efficient fine-tuning. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 37, pp. 12799–12807.

Gallegos, I.O., coauthors, 2024. Bias and fairness in large language models: A survey. Comput. Linguist..

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H., Wang, H., 2023, Retrieval-augmented generation for large language models: A survey, 2, (1) arXiv preprint arXiv:2312.10997.

Ge, Q., Qi, H.J., Dunn, M.L., 2013. Active materials by four-dimension printing. Appl. Phys. Lett. 103 (13).

Ghanem, H., Cruz, C., 2024. Fine-tuning vs. prompting: evaluating the knowledge graph construction with LLMs. In: 3rd International Workshop on Knowledge Graph Generation from Text (Text2KG) Co-Located with the Extended Semantic Web Conference. ESWC 2024, Vol. 3747, p. 7.

Ghedini, E., Hashibon, A., Friis, J., Goldbeck, G., Schmitz, G., De Baas, A., 2017. EMMO the European materials modelling ontology. In: EMMC Workshop on Interoperability in Materials Modelling. St John's Innovation Centre Cambridge, pp. 7–8.

Han, Z., Gao, C., Liu, J., Zhang, J., Zhang, S.Q., 2024. Parameter-efficient fine-tuning for large models: A comprehensive survey. Trans. Mach. Learn. Res. Published version; originally arXiv:2403.14608.

Hastings, J., Jeliazkova, N., Owen, G., Tsiliki, G., Munteanu, C.R., Steinbeck, C., Willighagen, E., 2015. eNanoMapper: harnessing ontologies to enable data integration for nanomaterial risk assessment. J. Biomed. Semant. 6, 1–15.

Hinton, T.J., Jallerat, Q., Palchesko, R.N., Park, J.W., Grodzicki, M.S., Shue, H.-J., Ramadan, M.H., Hudson, A.R., Feinberg, A.W., 2015. 3D printing PDMS elastomer in a hydrophilic support bath via freeform reversible embedding. Sci. Adv. 1 (9), e1500758. http://dx.doi.org/10.1126/sciadv.1500758.

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L.A., Welbl, J., Clark, A., et al., 2022. Training compute-optimal large language models. In: Advances in Neural Information Processing Systems. Vol. 35, URL: https://proceedings.neurips.cc/paper_files/paper/2022/file/c1e2faff6f588870935f114ebe04a3e5-Paper-Conference.pdf.

Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., de Melo, G., Gutierrez, C., Kirrane, S., Labra Gayo, J.E., Navigli, R., Neumaier, S., Ngonga Ngomo, A.-C., Polleres, A., Rashid, S.M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., Zimmermann, A., 2022. Knowledge graphs. In: Synthesis Collection of Technology, Springer, http://dx.doi.org/10.1007/978-3-031-01918-0.

Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2022. LoRA: Low-rank adaptation of large language models. In: International Conference on Learning Representations. ICLR, URL: https://openreview.net/forum?id=nZeVKeeFYf9.

HydrogelDesign, 2023. Hydrogel design tools. URL: https://hydrogeldesign.org/tools/.

Ihsan, A.Z., Dessì, D., Alam, M., Sack, H., Sandfeld, S., 2021. Steps towards a dislocation ontology for crystalline materials. arXiv preprint arXiv:2106.15136.

Jain, A., Ong, S.P., Hautier, G., Chen, W., Richards, W.D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., et al., 2013. The materials project: A materials genome approach to accelerating materials innovation. http://dx.doi.org/10.1063/1.4812323.

Jee, H., Witherell, P., 2017. A method for modularity in design rules for additive manufacturing. Rapid Prototyp. J. 23 (6), 1107–1118.

Jeong, C., 2024. Domain-specialized LLM: Financial fine-tuning and utilization method using Mistral 7B. J. Intell. Inf. Syst. 30 (1), 93–120. http://dx.doi.org/10.13088/jiis.2024.30.1.093, URL: http://dx.doi.org/10.13088/jiis.2024.30.1.093.

Ji, Z., Lee, N., Fries, J., Liang, P., Yang, T., 2023. Survey of hallucination in natural language generation. ACM Comput. Surv. http://dx.doi.org/10.1145/3571730.

Jiang, Z.-C., Liu, Q., Xiao, Y.-Y., Zhao, Y., 2024. Liquid crystal elastomers for actuation: A perspective on structure-property-function relation. Prog. Polym. Sci. 153, 101829. http://dx.doi.org/10.1016/j.progpolymsci.2024.101829.

Jiang, A.Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D.S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L.R., Lachaux, M.-A., Stock, P., Scao, T.L., Lavril, T., Wang, T., Lacroix, T., Sayed, W.E., 2023. Mistral 7B. arXiv, arXiv:2310.06825 URL: https://arxiv.org/abs/2310.06825.

Jing, Y., Zhou, G., Zhang, C., Chang, F., Li, J., 2025. Knowledge graph construction with meta-learning for continuously accumulated manufacturing knowledge. Comput. Ind. 172, 104353. http://dx.doi.org/10.1016/j.compind.2025.104353.

Katzer, B., Klinder, S., Schulz, K., 2025. Towards an automated workflow in materials science for combining multi-modal simulation and experimental information using data mining and large language models. Mater. Today Commun. 45, 112186. http://dx.doi.org/10.1016/j.mtcomm.2025.112186.

Kayusi, F., González Vallejo, R., Juma, L., Omwenga, M.K., Chavula, P., 2025. A bibliometric analysis of segment anything (SA) research: Global trends, key contributors, and thematic insights. Babylon. J. Mach. Learn. http://dx.doi.org/10.58496/BJML/2025/003, URL: https://mesopotamian.press/journals/index.php/BJML/article/view/754.

Khalighinejad, G., Scott, S., Liu, O., Anderson, K.L., Stureborg, R., Tyagi, A., Dhingra, B., 2025. MatViX: Multimodal information extraction from visually rich articles. In: Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics. NAACL 2025.

Kim, S., Joo, S.J., Kim, D., Jang, J., Ye, S., Shin, J., Seo, M., 2023. The CoT collection: Improving zero-shot and few-shot learning of language models via chain-of-thought fine-tuning. arXiv, arXiv:2305.14045 URL: https://arxiv.org/abs/2305.14045.

Kuenneth, C., Ramprasad, R., 2022. polyOne data set – 100 million hypothetical polymers including 29 properties. http://dx.doi.org/10.5281/zenodo.7124188.

Lambrix, P., Armiento, R., Li, H., Hartig, O., Pour, M.A.N., Li, Y., 2024. The materials design ontology. Semant. Web 15 (2), 481–515. http://dx.doi.org/10.3233/SW-233340.

Lee, B., Ahmed-Kristensen, S., 2025. D3 framework: An evidence-based data-driven design framework for new product service development. Comput. Ind. 164, 104206. http://dx.doi.org/10.1016/j.compind.2024.104206.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al., 2020. Retrieval-augmented generation for knowledge-intensive NLP. In: NeurIPS.

Li, H., 2018. Deep learning for natural language processing: advantages and challenges. Natl. Sci. Rev. 5 (1), 24–26.

Li, H., Armiento, R., Lambrix, P., 2020. An ontology for the materials design domain. In: International Semantic Web Conference. Springer, pp. 212–227.

Li, M., Gholami, F., Yue, L., Fratarcangeli, M.R., Black, E., Shimokawa, S., Nomura, T., Tanaka, M., Kobayashi, H., Song, Y., et al., 2024. Coaxial-spun hollow liquid crystal elastomer fiber as a versatile platform for functional composites. Adv. Funct. Mater. 34 (42), 2406847.

Li, Y., Hui, B., Xia, X., Yang, J., Yang, M., Zhang, L., Si, S., Chen, L.-H., Liu, J., Liu, T., Huang, F., Li, Y., 2024. One-shot learning as instruction data prospector for large language models. arXiv, arXiv:2312.10302 URL: https://arxiv.org/abs/2312.10302.

Li, Y., Li, B., 2022. Direct ink writing 3D printing of polydimethylsiloxane-based soft and composite materials: a mini review. Oxf. Open Mater. Sci. 2 (1), itac008. http://dx.doi.org/10.1093/oxfmat/itac008.

Lialin, V., Deshpande, V., Rumshisky, A., 2023. Scaling down to scale up: A guide to parameter-efficient fine-tuning. arXiv preprint arXiv:2303.15647.

Liu, H., Tam, D., Muqeeth, M., Mohta, J., Huang, T., Bansal, M., Raffel, C.A., 2022. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. Adv. Neural Inf. Process. Syst. 35, 1950–1965.

Lu, Y., Wang, J., 2025. KARMA: Leveraging multi-agent LLMs for automated knowledge graph enrichment. arXiv:2502.06472 URL: https://arxiv.org/abs/2502.06472.

Ludwig, H., Schmidt, T., Kühn, M., 2025. An ontology-based retrieval augmented generation procedure for a voice-controlled maintenance assistant. Comput. Ind. 169, 104289. http://dx.doi.org/10.1016/j.compind.2025.104289.

Maedche, A., Staab, S., 2001. Ontology learning for the semantic web. IEEE Intell. Syst. 16 (2), 72–79. http://dx.doi.org/10.1109/5254.920602.

McDonald, J.C., Whitesides, G.M., 2002. Poly(dimethylsiloxane) as a material for fabricating microfluidic devices. Acc. Chem. Res. 35 (7), 491–499. http://dx.doi.org/10.1021/ar010110q.

McGuinness, D.L., Van Harmelen, F., et al., 2004. OWL web ontology language overview. W3C Recomm. 10 (10), 2004.

Merchant, A., Batzner, S., Schoenholz, S.S., Aykol, M., Cheon, G., Cubuk, E.D., 2023. Scaling deep learning for materials discovery. Nature 624 (7990), 80–85.

NASA, 2025. NASA shape memory repository. https://shapememory.grc.nasa.gov/.

Nlm-atics, 2024. Llmsherpa: Developer APIs to accelerate LLM projects. GitHub repository https://github.com/nlmatics/llmsherpa.

Oyekan, J., Turner, C., Bax, M., Graf, E., 2025. From ontologies to knowledge augmented large language models for automation: A decision-making guidance for achieving human–robot collaboration in industry 5.0. Comput. Ind. 171, 104329. http://dx.doi.org/10.1016/j.compind.2025.104329.

Pan, J.Z., Razniewski, S., Kalo, J.-C., Singhania, S., Chen, J., Dietze, S., Jabeen, H., Omeliyanenko, J., Zhang, W., Lissandrini, M., Biswas, R., de Melo, G., Bonifati, A., Vakaj, E., Dragoni, M., Graux, D., 2023. Large language models and knowledge graphs: Opportunities and challenges. arXiv preprint arXiv:2308.06374.

Parthasarathy, V.B., Zafar, A., Khan, A., Shahid, A., 2024. The ultimate guide to fine-tuning LLMs from basics to breakthroughs: An exhaustive review of technologies, research, best practices, applied research challenges and opportunities. arXiv arXiv:2408.13296 URL: https://arxiv.org/abs/2408.13296.

Peng, B., Li, C., He, P., Galley, M., Gao, J., 2023. Instruction tuning with GPT-4. arXiv preprint arXiv:2304.03277.

Peng, X., Wu, S., Sun, X., Yue, L., Montgomery, S.M., Demoly, F., Zhou, K., Zhao, R.R., Qi, H.J., 2022. 4D printing of freestanding liquid crystal elastomers via hybrid additive manufacturing. Adv. Mater. 34 (39), 2204890.

Piane, F.L., Baldoni, M., Gaspari, M., Mercuri, F., 2021. Introducing MAMBO: Materials and molecules basic ontology. arXiv preprint arXiv:2111.02482.

Polak, M.P., Morgan, D., 2025. Leveraging vision capabilities of multimodal LLMs for automated data extraction from plots. arXiv preprint arXiv:2503.12326.

Pour, M.A.N., Li, H., Armiento, R., Lambrix, P., 2023. Phrase2Onto: A tool to support ontology extension. Procedia Comput. Sci. 225, 1415–1424.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al., 2019. Language models are unsupervised multitask learners. OpenAI Blog 1 (8), 9.

ResearchRabbit, 2025. ResearchRabbit. https://www.researchrabbit.ai/.

Richbourg, N.R., Peppas, N.A., 2023. Structurally decoupled stiffness and solute transport in multi-arm poly (ethylene glycol) hydrogels. Biomaterials 301, 122272.

Richbourg, N., Wancura, M., Gilchrist, A., Toubbeh, S., Harley, B., Cosgriff-Hernandez, E., Peppas, N., 2021. Precise control of synthetic hydrogel network structure via linear, independent synthesis-swelling relationships. Sci. Adv. 7 (7), eabe3245.

Ruiz, E., Torres, M.I., del Pozo, A., 2023. Question answering models for human–machine interaction in the manufacturing industry. Comput. Ind. 151, 103988. http://dx.doi.org/10.1016/j.compind.2023.103988.

Saal, J.E., Kirklin, S., Aykol, M., Meredig, B., Wolverton, C., 2013. Materials design and discovery with high-throughput density functional theory: the open quantum materials database (OQMD). Jom 65, 1501–1509.

Saeedizade, M.J., Blomqvist, E., 2024. Navigating ontology development with large language models. In: European Semantic Web Conference. Springer, pp. 143–161.

Smith, B., 1996. Mereotopology: A theory of parts and boundaries. Data Knowl. Eng. 20 (3), 287–303. http://dx.doi.org/10.1016/S0169-023X(96)00015-8.

Solórzano-Requejo, W., Aguilar Vega, C., Zapata Martínez, R., Bodaghi, M., Díaz Lantada, A., 2025. Ontology for smart 4D printed material systems and structures synergically applied with generative artificial intelligence for creativity promotion. Smart Mater. Struct. 34 (1), 015045.

Su, C., Han, Y., Tang, X., Jiang, Q., Wang, T., He, Q., 2024. Knowledge-based digital twin system: Using a knowledge-driven approach for manufacturing process modeling. Comput. Ind. 159–160, 104101. http://dx.doi.org/10.1016/j.compind.2024.104101.

Takahashi, K.-i., Mamitsuka, H., Tosaka, M., Zhu, N., Yamago, S., 2024. CoPolDB: a copolymerization database for radical polymerization. Polym. Chem. 15 (10), 965–971.

Team, G., Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A.M., Hauth, A., Millican, K., et al., 2023. Gemini: a family of highly capable multimodal models. arXiv preprint arXiv:2312.11805.

Thomas, D.G., Pappu, R.V., Baker, N.A., 2011. NanoParticle ontology for cancer nanotechnology research. J. Biomed. Informatics 44 (1), 59–74.

Tibbits, S., 2013. The emergence of "4D printing". In: TED Conference.

Trewartha, A., Walker, N., Huo, H., Lee, S., Cruse, K., Dagdelen, J., Dunn, A., Persson, K.A., Ceder, G., Jain, A., 2022. Quantifying the advantage of domain-specific pre-training on named entity recognition tasks in materials science. Patterns 3 (4).

Trirat, P., Jeong, W., Hwang, S.J., 2024. AutoML-Agent: A multi-agent LLM framework for full-pipeline AutoML. arXiv:2410.02958 URL: https://arxiv.org/abs/2410.02958.

Ucar, T., Gonzalez-Martin, A., Lee, M., Szwarc, A.D., 2020. One-shot learning for language modelling. arXiv, arXiv:2007.09679 URL: https://arxiv.org/abs/2007.09679.

University of Chicago, 2023. Polymer property predictor and database (PPPDB). URL: https://pppdb.uchicago.edu/.

Val-Calvo, M., Aranguren, M.E., Mulero-Hernández, J., Almagro-Hernández, G., Deshmukh, P., Bernabé-Díaz, J.A., Espinoza-Arias, P., Sánchez-Fernández, J.L., Mueller, J., Fernández-Breis, J.T., 2025. OntoGenix: Leveraging large language models for enhanced ontology engineering from datasets. Inf. Process. Manage. 62 (3), 104042.

Vaswani, A., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst..

Venugopal, V., Olivetti, E., 2024. MatKG: An autonomously generated knowledge graph in material science. Sci. Data 11 (1), 217.

Vsevolodovna, R.I.M., Monti, M., 2025. Enhancing large language models through neuro-symbolic integration and ontological reasoning. arXiv:2504.07640 URL: https://arxiv.org/abs/2504.07640.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D., et al., 2022. Chain-of-thought prompting elicits reasoning in large language models. Adv. Neural Inf. Process. Syst. 35, 24824–24837.

Wu, G., Ling, C., Graetz, I., Zhao, L., 2024. Ontology extension by online clustering with large language model agents. Front. Big Data 7, 1463543.

Xu, B., Poo, M.-m., 2023. Large language models and brain-inspired general intelligence. Natl. Sci. Rev. 10 (10), nwad267.

Xu, L., Xie, H., Qin, S.-Z.J., Tao, X., Wang, F.L., 2023. Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment. arXiv preprint arXiv:2312.12148.

Yang, Z., Gan, Z., Wang, J., Hu, X., Lu, Y., Liu, Z., Wang, L., 2022. An empirical study of GPT-3 for few-shot knowledge-based VQA. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 36, pp. 3081–3089.

Yang, Y., Peng, Q., Wang, J., Zhang, W., 2024. Multi-LLM-agent systems: Techniques and business perspectives. arXiv:2411.14033, URL: https://arxiv.org/abs/2411.14033.

Yin, S., Fu, C., Zhao, S., Li, K., Sun, X., Xu, T., Chen, E., 2024. A survey on multimodal large language models. Natl. Sci. Rev. 11 (12), nwae403. http://dx.doi.org/10.1093/nsr/nwae403.

Zhang, X., Hu, C., Li, H., 2009. Semantic query on materials data based on mapping MATML to an owl ontology. Data Sci. J. 8, 1–17.

Zhang, X., Pan, D., Zhao, C., Li, K., 2016. MMOY: Towards deriving a metallic materials ontology from yago. Adv. Eng. Informatics 30 (4), 687–702.

Zhang, Y., Sadler, T., Taesiri, M.R., Xu, W., Reformat, M., 2024. Fine-tuning language models for triple extraction with data augmentation. In: Proceedings of the 1st Workshop on Knowledge Graphs and Large Language Models (KaLLM 2024). pp. 116–124.

Zhao, W.X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., et al., 2023. A survey of large language models. arXiv preprint arXiv:2303.18223.